Higher Algebra I

MATH 570 · Scribed Course Notes · Part 1: Free Groups, Presentations and Cayley Complexes

Definitions Results
Examples Notes
green text marks a term or notation at the point where it is defined

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1 Course Overview

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Note 1. The abbreviations used in the manuscript have been expanded in this edition.

Note 2. Text set in blue (or red) reproduces the scribe's own annotations, written in blue (or red) ink in the manuscript. Pink and orange shading in figures reproduces the highlighter used in the manuscript.

Note 3. The marker "Source manuscript page n" indicates where page n of the handwritten manuscript begins.

Note 4. Following the colour convention of the lecturer's notes, a term or a piece of notation is set in green at the point where it is defined, e.g. "a free group is ..." or "$F_S := \dots$". Italics are reserved for ordinary emphasis.

Note 5. The manuscript closes most definitions, examples and remarks with the mark ☒ and most proofs with □; this edition renders both as □.

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2 Free Groups

2.1 Words

Let S be a set.

Definition

A word is an n-tuple; n is the length of the word. □

Example

$S = \{a, b\}$, $w = (a, a, b, a)$, a 4-tuple. We usually denote it by $aaba$. □

Definition

e = the empty word. The set of all words is

$$W_S := \bigcup_{n \geq 0} S^n, \qquad S^0 := \{e\} \text{ by definition.}$$

W_S is called a free monoid on S, with multiplication defined as concatenation of words. A monoid is a set with an associative multiplication and a unit element e. □

2.2 Adding the Inverses

Let $S^{-1} := \{s^{-1} : s \in S\}$ be a set disjoint from S. Consider $W_{S_\pm}$, which is a free monoid on $S_\pm = S \sqcup S^{-1}$. It is not a group, as $aa^{-1} \neq e$.

Definition

A word is reduced if it does not contain ss^{-1} for any $s \in S_{\pm}$. □

Definition

$F_S :=$ the set of all reduced words in $W_{S_{\pm}}$. Define a multiplication on F_S in two steps:

(1) concatenation;
(2) reduction (i.e. removing bb^{-1} pairs to e).

□

Lemma

Every word in $W_{S_{\pm}}$ can be put in a unique reduced form, no matter the order of reduction.

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Proof. Obvious. ("Obvious.") □

Algorithm.

(1) Start from the left and reduce any letter that is followed by its inverse.
(2) Repeat (1) until the word is in its reduced form.

Example

$abb^{-1}a^{-1} \to aa^{-1} \to e$. □

Definition

F_S is called the (unique) free group on S. Heuristically, a free group is a group with an "independent" generating set S (in the sense that the generators have no relation, nor elements of finite order). □

Proposition

$W_S \hookrightarrow F_S$ by a monoid homomorphism.

Proof. Notice that every word in W_S is already reduced (no "aa^{-1}"). □

2.3 The Cayley Tree of F_S

Definition

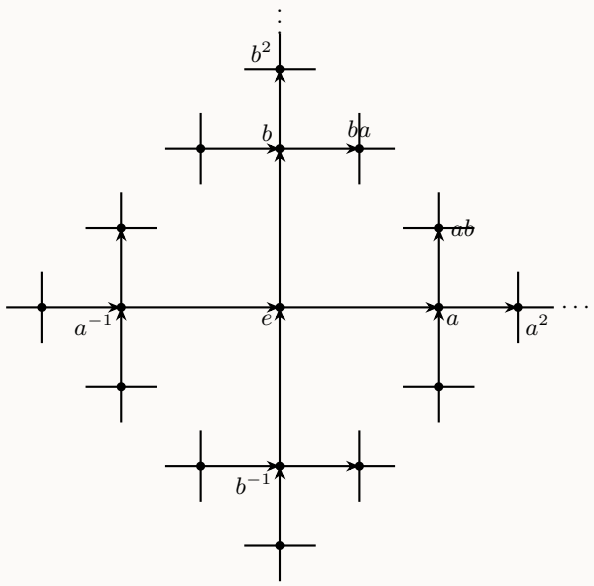
Let G be a group and $S \subseteq G$. $\mathrm{Cay}(G, S)$, the Cayley graph, is the graph with vertices $V = G$ and edges $E = \{(g, gs) : s \in S\}$, given by right multiplication in G:



□

Example

$S = \{a, b\}$. Reading a word on a Cayley graph = a walk on the edges.



□

Proposition

$\mathrm{Cay}(F_S, S)$ is a tree.

Proof. Suppose a cycle exists at a vertex g. Then translating it by g^{-1} gives a cycle at e. Yet every vertex is a word in F_S. Then the walk following this cycle

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gives a nonempty reduced word $= e$, which is impossible in F_S. □

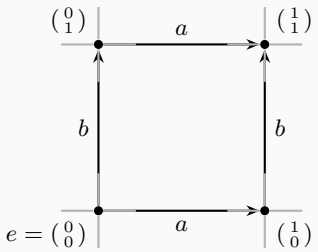
Remark. The converse holds: for $S \subseteq G$, $\mathrm{Cay}(G, S)$ is a tree $\iff$ G is free on S. □

Proposition

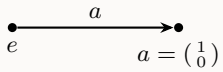
$\mathrm{Cay}(G, S)$ is connected $\iff$ S generates G, i.e. $G = \langle S \rangle$. □

Example

$G = \mathbb{Z}^2$. $S_1 = \left\{a = \binom{1}{0},\ b = \binom{0}{1}\right\}$ generates G, whereas $S_2 = \left\{\binom{1}{0}\right\}$ doesn't reach $\binom{0}{1}$, $\binom{1}{1}$:



vs. $S_2 = \left\{\binom{1}{0}\right\}$:



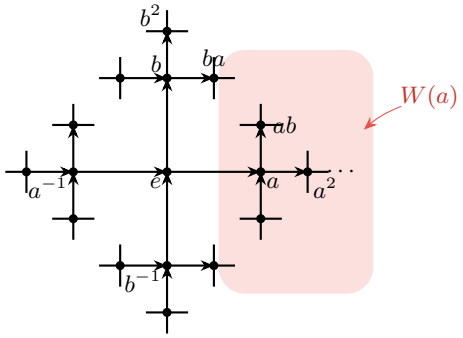
□

On the other hand, notice that there is a cycle in the Cayley graph, hence $\mathrm{Cay}(\mathbb{Z}^2, S_1)$ is not a tree $\implies$ $\mathbb{Z}^2$ is not free on S_1. The nontrivial relation between the generators is $\binom{1}{0} + \binom{0}{1} = \binom{0}{1} + \binom{1}{0}$. □

Proposition

$W_{S_\pm}$ = paths in $\mathrm{Cay}(F_S, S)$ starting from e.
F_S = all reduced forms in $W_{S_\pm}$ = paths with no backtracking (geodesic paths).
Reduction = deleting backtracks. □

Consider $W(a)$:= the set of reduced words starting from a.



Notice that $a\left(F_S \setminus W(a^{-1})\right) = W(a)$, i.e. three branches in the Cayley tree collapse to one. This makes it impossible to define an invariant mean (i.e. a finitely additive probability measure) on F_2

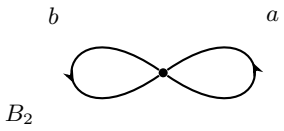
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(called non-amenability). (Although, this holds if $|S| \geq 2$; F_1 is indeed amenable.) This also leads to the Ping-Pong lemma. □

Fact

F_S acts on $\mathrm{Cay}(F_S, S)$ by $g \cdot (h, hs) = (gh, ghs)$. In general, a group G acts on a graph by bijections $g\colon V \to V$ respecting the edges: $(u, v) \in E \implies (gu, gv) \in E$. This is indeed an action, since left multiplication by g and right multiplication by s commute as functions. □

Consider F_2, the free group on two generators. $F_2 \backslash \mathrm{Cay}(F_2, S) =: B_2$ is a new quotient graph, obtained by looking at the orbits of the action of F_2 on $\mathrm{Cay}(F_2, S)$. There is one orbit for the vertices as an F_2-set ($V = F_2$, and F_2 acts on itself transitively); there are two orbits for the edges (F_2 acts on edges by translation; thus the labels $\uparrow b$, $\to a$ are precisely the orbits):



Margin note: $F_1 = \pi_1(S^1)$, $F_n = \pi_1(B_n)$ — the algebraic topology approach to constructing a free subgroup. □

Definition

w is cyclically reduced if it is reduced and its last letter $\neq$ (first letter)$^{-1}$. □

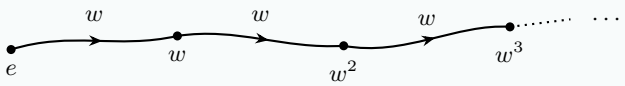
Example

$w = abab\,a^{-1}$ is not cyclically reduced. □

Proposition

If w is cyclically reduced, then $|w^k| = |k|\,|w|$. Moreover, if $w \neq e$,

$$\gamma = \bigcup_{k\in\mathbb{Z}} [w^k, w^{k+1}] \;=\; \text{concatenation of paths}$$

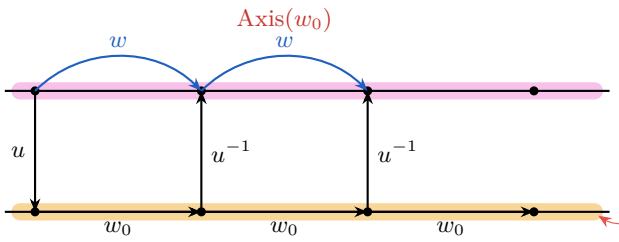


(no backtracking)

is a geodesic in $\mathrm{Cay}(G, S)$, called the axis of w, $\mathrm{Axis}(w)$. □

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Remark. If $w = uw_0u^{-1}$ with w_0 cyclically reduced, we can still define $\mathrm{Axis}(w) = u\,\mathrm{Axis}(w_0)$:

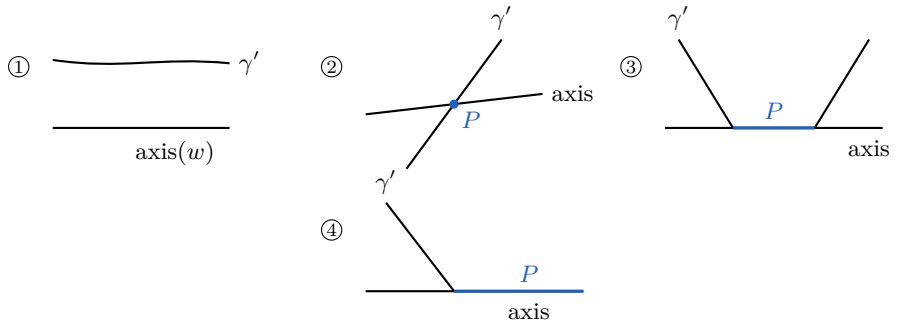


$w^k = uw_0^k u^{-1}$. □

Lemma

$\mathrm{Axis}(w)$ is the unique $\langle w\rangle$-invariant geodesic in the tree.

Proof. Let γ' be a geodesic which is $\langle w\rangle$-invariant. Assume w is cyclically reduced for simplicity. There are two possibilities, $\gamma' \cap \mathrm{axis}(w) = \varnothing$ or $\gamma' \cap \mathrm{axis}(w) \neq \varnothing$:



① Let $P :=$ the unique shortest path from γ' to the axis.

Claim: If both γ' and axis(w) are $\langle w \rangle$-invariant, then so is P, in all four cases. Yet P is not invariant unless $P = \text{axis}(w)$, since it will be translated along axis(w). □

Remark. Many elements can have the same axis. Hence the

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converse of the previous lemma is not true. See the exercise on pages 57–58. □

Exercise

$F_2 = \{e\} \sqcup W(a) \sqcup W(b) \sqcup W(a^{-1}) \sqcup W(b^{-1})$. (Obvious.) Show that

$$F_2 = W(a) \sqcup aW(a^{-1}) = W(b) \sqcup bW(b^{-1}).$$

Exercise (Paradoxical F_2)

Let $A \subseteq F_2$. Want: an invariant mean $\mu \colon 2^{F_2} \to [0,1]$ such that

$$\mu(\varnothing) = 0, \qquad \mu(A \sqcup B) = \mu(A) + \mu(B), \qquad \mu(F_2) = 1, \qquad \mu(sA) = \mu(A).$$

Show that there is no invariant mean on F_2.

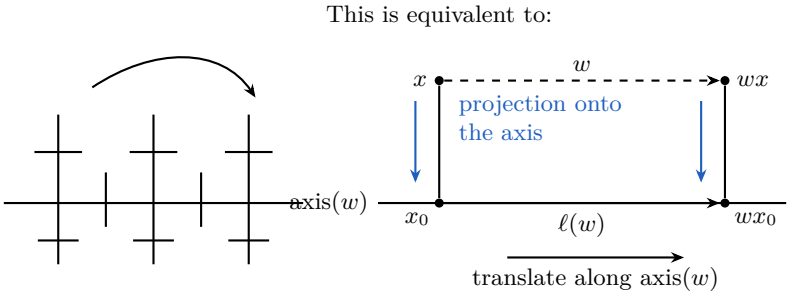
Remark. An invariant probability measure on a group = a function satisfying the axioms above & countable additivity, & potentially defined on a σ-algebra of G (a subset of 2^{F_2} closed under countable union and complement). □

Solution. Suppose such an invariant mean μ exists. Then

$$\begin{aligned} 1 &= \mu(W(a)) + \mu(W(a^{-1})) + \mu(\{e\}) + \mu(W(b)) + \mu(W(b^{-1})) \\ &\geq \mu(W(a)) + \mu(aW(a^{-1})) + \mu(W(b)) + \mu(bW(b^{-1})) \\ &= \mu(F_2) + \mu(F_2) = 1 + 1 = 2. \qquad \times \end{aligned}$$

□

2.4 Translation by the Action



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This path is a geodesic, by observing that (x_0, x) is translated to (wx_0, wx).

Definition

We call the length of the translation by (x_0, wx_0) the translation length, denoted by $\ell(w) = |w|$. In general, for a non-cyclically-reduced word $w = uw_0u^{-1}$ with w_0 cyclically reduced, $\ell(w) = |w_0|$. □

Definition

A length on a group is a map $\ell\colon G \to \mathbb{N}$ satisfying

(i) $\ell(e) = 0$;
(ii) $\ell(s) = \ell(s^{-1})$;
(iii) $\ell(st) \le \ell(s) + \ell(t)$.

□

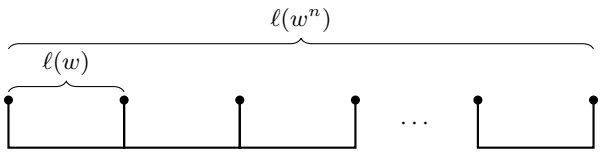
Remark. Our ℓ is conjugacy invariant: $\ell(uw_0u^{-1}) = \ell(w_0)$; while it does not satisfy (iii): take $s = aba^{-1}$, $t = b$. □

Proposition

$\ell(w^n) = |n|\,\ell(w)$ for all $w \in F_S$, and

$$\ell(w) = \lim_{n\to\infty} \frac{|w^n|}{n} = \lim_{n\to\infty} \frac{|w_0^n|}{n}, \qquad \text{if } w = uw_0u^{-1}.$$

Proof. The first one is obvious by a drawing:



For the second, notice that

$$|w^n| = |uw_0^nu^{-1}| = 2|u| + |w_0^n| = 2|u| + \ell(w_0^n) = 2|u| + |n|\,|w_0| \implies \frac{|w^n|}{|n|} = \frac{2|u|}{|n|} + |w_0|$$

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$\implies |w_0| = \lim_{n\to\infty}|w^n|/|n|$, provided the limit exists. □

Exercise (axes and translation lengths)

(1) Show $\operatorname{axis}(u) = \operatorname{axis}(u^n)$.
Hint: Recall the uniqueness of invariant geodesics.

(2) If $u, v \in F_S$ and $\operatorname{axis}(u) = \operatorname{axis}(v) = \gamma$ (not only translated-equivalent: we mean the *same* axis), then u, v generate a cyclic group.
Hint: Show $H = \operatorname{Stab}(\gamma)$ is a cyclic group by looking at

$$\varphi\colon H \longrightarrow \mathbb{Z}, \qquad s \longmapsto \text{the } \textit{signed} \text{ translation length of } s,$$

fixing an orientation. This is in fact a group homomorphism and $\ker\varphi = \{s : \ell(s) = 0\} = \{e\}$. Thus $\langle u, v\rangle \le H \cong \varphi(H) \le \mathbb{Z}$. Every subgroup of a cyclic group must be cyclic; $\mathbb{Z} = \langle 1\rangle$ is cyclic, hence so is $\langle u, v\rangle$ (and H).

(3) $u, v \in F_S$. If $u^n = v^m$, then u, v commute.
Proof: $\operatorname{Axis}(u) = \operatorname{Axis}(u^n) = \operatorname{Axis}(v^m) = \operatorname{Axis}(v) \implies \langle u, v\rangle$ cyclic.

(4) $u, v \in F_S$, $uv = vu \implies \langle u, v\rangle$ cyclic.
Proof: $uv = vu \implies vuv^{-1} = u \implies \operatorname{Axis}(vuv^{-1}) = \operatorname{Axis}(u) \implies v\operatorname{Axis}(u) \overset{*}{=} \operatorname{Axis}(vuv^{-1}) = \operatorname{Axis}(u)$ ($*$: exercise) $\implies \operatorname{Axis}(u)$ is $\langle v\rangle$-invariant $\implies \operatorname{Axis}(u) = \operatorname{Axis}(v)$ by uniqueness of the invariant geodesic.

(5) Suppose $uv^pu^{-1} = v^q$ (called the BS(p, q) (Baumslag–Solitar) relation),

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with $\boxed{v \neq e}$. Then $|p| = |q|$:

$$|q|\,\ell(v) = \ell(v^q) = \ell(uv^pu^{-1}) = \ell(v^p) = |p|\,\ell(v) \implies |p| = |q|,$$

i.e. no BS-relation exists in the free group unless $|p| = |q|$.
If $p = q$, then $uv^p = v^pu \implies \operatorname{Axis}(u) = \operatorname{Axis}(v^p) = \operatorname{Axis}(v) \implies \langle u, v\rangle$ cyclic (by claims 1, 2, 3).
If $p = -q$, then $uv^p = v^{-p}u \implies \langle u, v\rangle$ cyclic $\implies uv^p = v^{-p}u = uv^{-p} \implies v^{2p} = e$, impossible in a free group.
I.e. if a BS-relation exists in a free group with $v \neq e$, we must have $p = q$.

Corollary (6)

For any group homomorphism $\varphi\colon \mathbb{Z}^2 \to F_S$, $\operatorname{im}(\varphi)$ must be cyclic.

Proof. $\operatorname{im}(\varphi) = \varphi\big(\langle \binom{1}{0}, \binom{0}{1}\rangle\big) = \langle \varphi\binom{1}{0}, \varphi\binom{0}{1}\rangle$, and $\varphi\binom{1}{0}, \varphi\binom{0}{1}$ commute since $\binom{1}{0}, \binom{0}{1}$ commute in $\mathbb{Z}^2$. □

Exercise (7)

Recall that for a group G we define the center subgroup $Z(G) := \{s \in G : sa = as,\ \forall a \in G\}$. Then

$$Z(F_1) = \mathbb{Z}, \qquad Z(F_n) = \{e\},\ \forall n \ge 2.$$

Solution. F_1 has a generating set consisting of a single element a. Hence F_1 is cyclic and abelian, and $Z(F_1) = \langle a\rangle = \mathbb{Z}$.

Suppose $n \geq 2$. Suppose $S = \{s_1, \dots, s_n\}$ is the

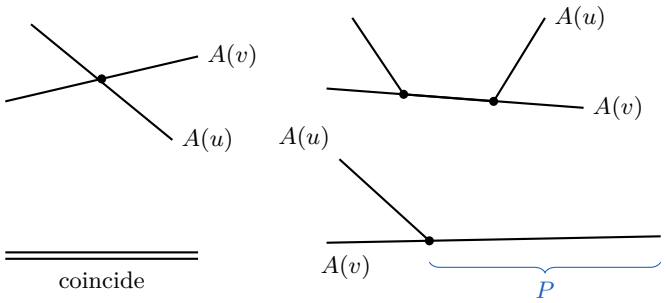
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letter set (which generates F_n). Let $w \neq e$, $w = s_i w_0$. Then $s_j w \neq w s_j$ for $j \neq i$. □

Exercise (8)

For u, v cyclically reduced and non-commuting, $\mathrm{Axis}(u) \cap \mathrm{Axis}(v)$ is a finite segment.

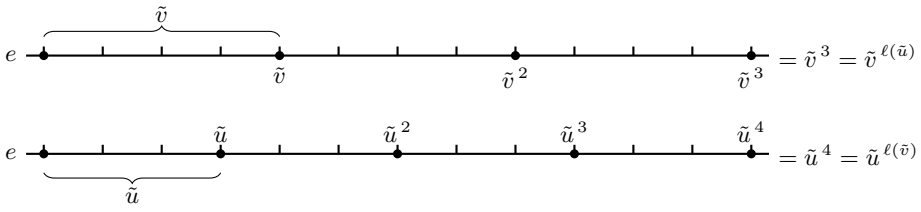
Solution. Since u, v are cyclically reduced, $\mathrm{Axis}(u) \cap \mathrm{Axis}(v) \ni e$, so it is $\neq \varnothing$. Recall the possibilities:



$\mathrm{Axis}(u) \neq \mathrm{Axis}(v)$ since u, v don't commute. We show that $\mathrm{Axis}(u) \cap \mathrm{Axis}(v)$ is not a ray. If it were, let $\tilde{u}, \tilde{v}$ be the proper choices of $u^{\pm 1}, v^{\pm 1}$ respectively so that $\tilde{u}, \tilde{v} \in P$. Then since each edge on the geodesic is a letter, powers of $\tilde{u}$ and $\tilde{v}$ will eventually meet:

$$\tilde{v}^{\,\ell(\tilde{u})} = \tilde{u}^{\,\ell(\tilde{v})}.$$

By (3), $\tilde{v}$ and $\tilde{u}$ commute, implying v, u commute. Contradiction.



□

Theorem (9)

For $u, v \in F_S$ with $\mathrm{Axis}(u) \cap \mathrm{Axis}(v) = \varnothing$, we have

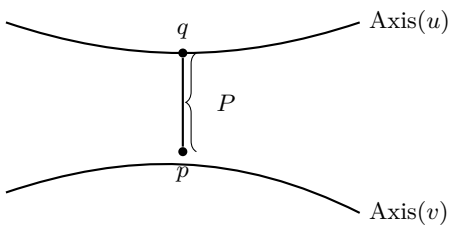
$$\ell(uv) = \ell(u) + \ell(v) + 2\, d\big(\mathrm{Axis}(u), \mathrm{Axis}(v)\big),$$

where d denotes

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the length of the unique (shortest) path from $\mathrm{Axis}(u)$ to $\mathrm{Axis}(v)$.

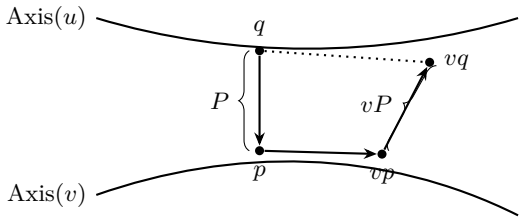
Solution. This is an important result in geometric group theory. Let $[p, q] = P$ be the unique geodesic from $\mathrm{Axis}(v)$ to $\mathrm{Axis}(u)$:



Let $g = uv$. In the pictures below, solid lines = paths in the tree; dashed lines = not a path in the tree.

Claim 1. $d(q, vq) = \ell(v) + 2d(p, q)$.

Note that $q \to p \to vp \to vq$ is *the* unique geodesic from q to vq: $p \to vp$ is a path (in fact, both points sit on Axis(v) since it is the only $\langle v \rangle$-invariant geodesic); $vp \to vq$ is a path since it is the path $p \to q$ translated by the action of v. Furthermore, there is no backtracking:

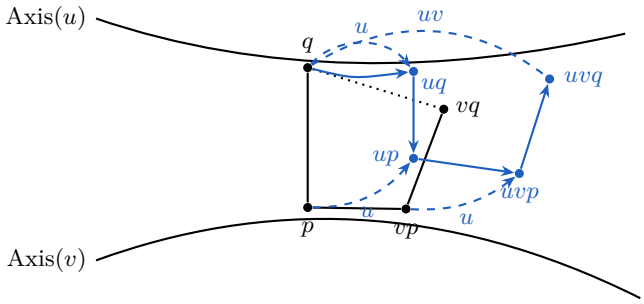


Lastly, $\ell(v) = |[p, vp]|$ since $[p, vp]$ sits on Axis(v). 1

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Claim 2. $d(q, uvq) = \ell(u) + \ell(v) + 2d(p, q)$.

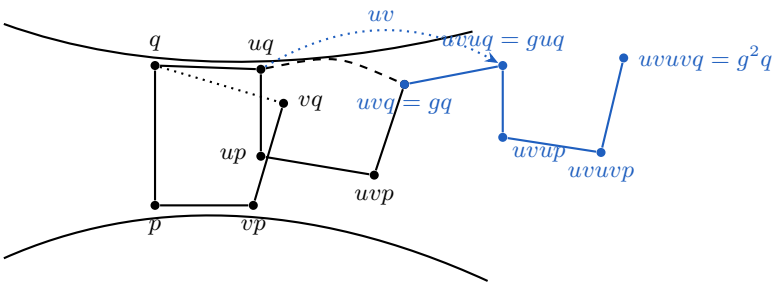
Notice that $[q, gq] = [q, uvq] = q \to uq \to up \to uvp \to uvq$. Each of them is a path with no backtracking in between (same logic as Claim 1):



Hence, following the geodesic,

$$\begin{aligned} d(q, gq) &= d(q, uq) + d(uq, up) + d(up, uvp) + d(uvp, uvq) \\ &= \ell(u) + d(p, q) + \ell(v) + d(p, q). \end{aligned}$$ 2

Claim 3. $q \in \text{Axis}(g)$. Consider the geodesic $\bigcup_{k \in \mathbb{Z}} [g^k q, g^{k+1} q]$. There is no backtracking at the contact:



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More precisely: $[uvq, uvuq] \subseteq uv\,\mathrm{Axis}(u)$ and $[uvp, uvq] \subseteq uvP$. Thus $uv\,\mathrm{Axis}(u) \cap uvP$ has an edge (a backtrack) $\iff$ $\mathrm{Axis}(u) \cap P$ has an edge, which is false by our assumption that P intersects $\mathrm{Axis}(u)$ and $\mathrm{Axis}(v)$ at a node (respectively). Hence $\bigcup_{k\in\mathbb{Z}}[g^k q, g^{k+1}q]$ is a bi-infinite $\langle g\rangle$-invariant geodesic, hence $\bigcup_{k\in\mathbb{Z}}[g^k q, g^{k+1}q] = \mathrm{Axis}(g)$, and $[q, gq] \subseteq \mathrm{Axis}(g)$. 3

Hence

$$\ell(uv) = \ell(g) = d(q, gq) = \ell(u) + \ell(v) + 2d(p, q) = \ell(u) + \ell(v) + 2d\big(\mathrm{Axis}(u), \mathrm{Axis}(v)\big). \qquad \square$$

2.5 Free Generating Sets / Bases

Notation (reduced form)

Denote $w = s_1^{r_1} \cdots s_n^{r_n}$ with $s_i \in S$, $r_i \in \mathbb{Z}$, $s_k \neq s_{k+1}$, $r_k \neq 0$; this expression is unique. $\square$

Proposition (Universal property of free groups)

Let $f_0 \colon S \to G$ be any map into a group. Then there exists a unique homomorphism $f \colon F_S \to G$ that extends f_0.

Proof (heuristic). Define

$$f(s_1^{r_1} \cdots s_n^{r_n}) := f_0(s_1)^{r_1} f_0(s_2)^{r_2} \cdots f_0(s_n)^{r_n}.$$

Then uniqueness follows, and one verifies that f is a homomorphism; in particular, one should verify that cancellation in F_S doesn't affect the product. $\square$

2.6 Free Bases (Nielsen)

Definition

A free generating set / (free) basis of a free group G is a set $S \subseteq G$ such that $\iota \colon S \hookrightarrow G$ extends to an isomorphism

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$F_S \to G$ by the universal property. $\square$

Proposition

$F_S \cong F_T$ (group isomorphism) $\iff$ $|S| = |T|$, for all $|S|, |T| < \infty$.

Proof. ($\Leftarrow$) Suppose $|S| = |T|$ and $f\colon S \xrightarrow{\sim} T$. Then we have $f\colon S \hookrightarrow F_T$ (extend the codomain) $\implies f\colon F_S \to F_T$ by the universal property. On the other hand, we get a homomorphism $g\colon F_T \to F_S$ in the same way. Notice $fg = \mathrm{Id}_{F_T}$, $gf = \mathrm{Id}_{F_S}$, hence $f^{-1} = g$ and f is an isomorphism.

($\Rightarrow$) Take $G = F_S$. Then $G^2 = \langle s^2 : s \in G\rangle$ is a normal subgroup of G. Then G/G^2 has the property $s^2 = e$ for all $s \in G/G^2$, so $s = s^{-1}$, $t = t^{-1}$, $sts^{-1}t^{-1} = stst = st(st)^{-1} = e$. Hence G/G^2 is abelian. Let $G = F_S$. Then G is finitely generated $\implies$ G/G^2 is finitely generated. Furthermore, G/G^2 has exponent 2 (i.e. all elements have order ≤ 2): it is a $(\mathbb{Z}/2\mathbb{Z})$-vector space, with addition being the group operation of G/G^2 and scalar multiplication being exponentiation. Let $n := |S|$.

Let $\exp_w(s_i) :=$ the sum of all exponents of s_i in w; for instance, $w = a^{-3}bc^2a^4$ has $\exp_w(a) = 1$, $\exp_w(b) = 1$. Define $\varphi\colon F_S \to (\mathbb{Z}/2\mathbb{Z})^n$ via

$$w \longmapsto \big(\exp_w(s_1), \exp_w(s_2), \dots, \exp_w(s_n)\big) \bmod 2,$$

a group homomorphism. Then $\underline{\ker\varphi = F_S^2}$:

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- $F_S^2 \subseteq \ker\varphi$: trivial, every element in F_S^2 has exponents $\equiv 0 \pmod 2$.
- $\ker\varphi \subseteq F_S^2$: let $w \in \ker\varphi$ and consider $[w] \in F_S/F_S^2$. Then since F_S/F_S^2 is abelian, we can rearrange the terms of w into w_0^2 for some $w_0 \in F_S$, since $w \in \ker\varphi$ has exponent sums which are multiples of 2. For instance, for $abb^{-3}a^5 \in \ker\varphi$,

$$abb^{-3}a^5\,(F_S^2) = a^3b^{-1}a^3b^{-1}\,F_S^2 = (a^3b^{-1})^2\,F_S^2 = F_S^2.$$

  Thus $[w] = F_S^2 \implies w \in F_S^2$.

Thus $F_S/F_S^2 \cong (\mathbb{Z}/2\mathbb{Z})^n$.

$$\begin{array}{ccccc} F_S & \longrightarrow & F_S/F_S^2 & = & H_1(F_S, \mathbb{Z}/2\mathbb{Z}) \\ \downarrow & \searrow^{\varphi} & \downarrow \cong & & \\ F_S/\ker\varphi & \xrightarrow[\cong]{} & (\mathbb{Z}/2\mathbb{Z})^n & & \end{array}$$

On the other hand, we can use $[G,G] = \langle sts^{-1}t^{-1} : s,t \in G\rangle$ as well. Note that $[G,G] \trianglelefteq G$ is normal:

$$w\,sts^{-1}t^{-1}\,w^{-1} = wsw^{-1}\,wtw^{-1}\,ws^{-1}w^{-1}\,wt^{-1}w^{-1} = [wsw^{-1}, wtw^{-1}].$$

Then $H_1(G,\mathbb{Z}) = G/[G,G]$ is abelian, called the abelianization of G. ($H_1(G,\mathbb{Z})$ is homology notation (MATH 571).) Then in fact, looking at the map $\varphi\colon F_S \to \mathbb{Z}^{|S|}$, $S = \{s_1, \dots, s_n\}$, defined via

$$w \longmapsto \big(\exp_w(s_1), \exp_w(s_2), \dots, \exp_w(s_n)\big),$$

it is a homomorphism. Then $\ker\varphi = [G,G]$ in the same fashion as above, thus $F_S/[F_S,F_S] \cong \mathbb{Z}^{|S|}$. Since $F_S \cong F_T$, we have $F_S/F_S^2 \cong F_T/F_T^2$, or $F_S/[F_S,F_S] \cong F_T/[F_T,F_T]$. Thus $\mathbb{Z}^{|S|} \cong \mathbb{Z}^{|T|}$ as free $\mathbb{Z}$-modules, or $(\mathbb{Z}/2\mathbb{Z})^{|S|} \cong (\mathbb{Z}/2\mathbb{Z})^{|T|}$ as $(\mathbb{Z}/2\mathbb{Z})$-vector spaces.

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Either one lets us conclude $|S| = |T|$. □

Definition

Let G be a free group and S a basis. Then the rank of G is $\mathrm{rk}(G) := \#S$. This is well defined by the last proposition. □

Proposition

A generating set of size n of F_n is a basis.

Proof. Suppose the set of letters is $S = \{s_1, \dots, s_n\}$. Let $\{a_1, \dots, a_n\}$ be a generating set of F_n. Define $\varphi\colon S \to F_n$ by $s_i \mapsto a_i$. Extend the map to a unique homomorphism $F_S = F_n \to F_n$. This is surjective since the $\{a_i\}$ are generators. F_n is Hopfian (below) $\implies$ φ is injective. Consider the map $\rho\colon F_{\{a_i\}} \to F_{\{s_i\}}$ that relabels $a_i \mapsto s_i$, with the a_i in $F_{\{a_i\}}$ viewed as letters with no relations. This is a well-defined isomorphism. Then

$$\varphi \circ \rho\colon F_{\{a_i\}} \xrightarrow{\rho} F_{\{s_i\}} \xrightarrow{\varphi} F_n$$

is an isomorphism extending $\iota\colon \{a_i\} \hookrightarrow F_n$. Therefore, by definition, $\{a_i\}$ is a basis of F_n. □

Definition

G is Hopfian if every surjective homomorphism $f\colon G \twoheadrightarrow G$ is injective. Equivalently, for all $N \trianglelefteq G$, $G/N \cong G \iff N = \{e\}$. □

Example

$\{a, b\}$, $\{a, a^n b\}$, $\{a, a^n b a^{-n}\}$ are all bases of F_2, for all $n \in \mathbb{N}$. □

Corollary

Let S be a generating set of F_n. Then $\#S \geq n$.

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Proof. Consider the surjective homomorphisms

$$F_n \twoheadrightarrow \mathbb{Z}^n \twoheadrightarrow (\mathbb{Z}/2\mathbb{Z})^n.$$

Then a generating set of F_n of size $< n$ $\implies$ a generating set of $\mathbb{Z}^n$ of size $< n$ $\implies$ a generating set of $(\mathbb{Z}/2\mathbb{Z})^n$ of size $< n$. Yet this is impossible for vector spaces, as the bases are the *minimal* spanning sets.

Note that this is the same map we used when proving that bases of free modules have the same cardinality: surjecting onto vector spaces of the same rank (dimension). □

Example

Finite groups are Hopfian. □

Proposition

$\mathbb{Z}^n$ and F_2 are Hopfian (not true for $\mathbb{Z}^\infty$, F_∞).

Definition

A group G is residually finite if for all $s \neq e$, $s \in G$, there exists a homomorphism $f\colon G \to H$ with $|H| < \infty$ and $f(s) \neq e$. □

Example

$\mathbb{Z}$ is residually finite: for all $n \in \mathbb{Z}$ there exists $f\colon \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$ with $f(n) \neq 0$, for $m > |n|$ (as long as $m \nmid n$). □

Example

$f\colon F_2 \to \mathbb{Z}/m\mathbb{Z}$. For $w = a^n$: $a \mapsto \bar{1}$, $b \mapsto \bar{0}$.

$$\begin{array}{ccc} F_2 & \xrightarrow{f} & \mathbb{Z}/m\mathbb{Z} \\ \downarrow & & \uparrow \\ \mathbb{Z}^2 & \longrightarrow & (\mathbb{Z}/m\mathbb{Z})^2 \end{array}$$

Alternatively, the map can be viewed as abelianization, then reducing modulo m, then projecting onto

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the first coordinate. □

Example

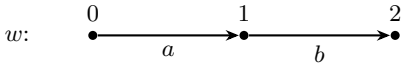
$\mathbb{Z}^n$; $\mathbb{Z}^\infty = \bigoplus_{i=1}^\infty \mathbb{Z} := \left\{\{n_i\}_{i=1}^\infty : n_i \in \mathbb{Z},\ n_i = 0\ \forall^\infty i\right\}$, i.e. sequences of integers that are 0 for all but finitely many indices; $\prod_{i=1}^\infty \mathbb{Z} = \mathbb{Z}^{\mathbb{N}} = \{\text{sequences of integers}\}$ — are all residually finite. □

Proposition

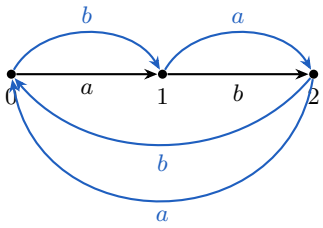
F_S is residually finite, for all $|S| < \infty$.

Proof (Hall, Stallings). We describe the general approach with F_S, $w = a_1 a_2 \cdots a_n$, accompanied by examples in F_2 with $w = ab$. Suppose $S = \{s_1, \ldots, s_n\}$ is the letter set.

Step 1: Draw out w to the right as a straight line.



Step 2: Complete the graph so that every node has exactly one in-s_i-edge and one out-s_i-edge, for all $s_i \in S$. Note that this completion is not unique.





This is possible, as each s_i-edge gives exactly one in-s_i-edge and one out-s_i-edge to some vertices in the graph. (If one misses an in-s_i-edge, there must

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be another node missing an out-s_i-edge.)

Remark. In particular, consider the subgroup G of F_2 consisting of words that fix 0 in the graph above (e.g. $w_1 = aba$, $w_2 = a^3b^3$). Then G acts on the Cayley graph $\mathrm{Cay}(F_2, \{a, b\})$ by multiplication on the left. In fact, the quotient graph we obtain as $G\backslash \mathrm{Cay}(F_2, \{a, b\})$ is exactly the graph above. This is also a covering space of the space B_2 (p. 5). □

Now, for $M = |w| + 1$, define a homomorphism $\varphi\colon F_S \to S_M^{\mathrm{op}}$ via $s_i \mapsto \sigma_i :=$ the permutation of the vertices given by walking along the s_i-edges, where S_M^{op} is the symmetric group with the order of multiplication reversed, to make the homomorphism preserve multiplication (e.g. $(34)\,\mathrm{op}(125) := (125)(34)$). With F_2: $\varphi(a) = (0\,1\,2)$, $\varphi(b) = (0\,1\,2)$. In particular,

$$\varphi(w) = \varphi(a_1)\cdots\varphi(a_n) = (n{-}1\ \ n)\cdots(1\ 2)(0\ 1)$$

sends 0 to $n = |w|$. Thus $\varphi(w) \neq e$ in S_M^{op}. □

2.7 Another Characterization of Residual Finiteness

Exercise

H is of finite index in $G \implies H$ contains a finite index normal subgroup K.

Proof. Consider $G/H = \{s_1H, \ldots, s_nH\} = X$, $n = |G/H| = [G : H]$. Then G acts on X, giving a homomorphism $f\colon G \to S_n$. Hence

$$\ker f = \{g \in G : gs_iH = s_iH\} = \bigcap_{1\le i\le n} s_iHs_i^{-1} =: K = \bigcap_{s\in G} sHs^{-1}.$$

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Hence, by the first isomorphism theorem, $[G : K] = |\mathrm{Im}(f)| \le n!$

$$\implies [H : K] = \frac{[G : K]}{[G : H]} \le n!.$$ □

Corollary

Let G be a group and $H \le G$ with $[G : H] < \infty$. Then there exists $K \trianglelefteq G$, $K \le H$, such that for all $s \notin H$, the projection homomorphism $\varphi\colon G \twoheadrightarrow G/K$ maps $s \mapsto \varphi(s) \neq e$.

Proposition

A countable group G is residually finite $\iff$ there exist finite index subgroups $G \ge G_1 \ge G_2 \ge \cdots$ such that $\bigcap_{n\ge1} G_n = \{e\}$.

Proof. ($\Rightarrow$) Enumerate the nontrivial elements of the group: $G \setminus \{e\} = \{a_n\}_n$. Then for all a_n there exist H_n with $|H_n| < \infty$ and a homomorphism $\varphi_n\colon G \to H_n$ with $\varphi_n(a_n) \neq e$ in H_n. Then $\ker\varphi_n \trianglelefteq G$ and

$[G : \ker \varphi_n] \leq |H_n| < \infty$ is of finite index. Consider

$$G \geq \ker \varphi_1 \geq \ker \varphi_1 \cap \ker \varphi_2 \geq \cdots \geq \bigcap_{i \leq n} \ker \varphi_i \geq \cdots.$$

Then $\left[G : \bigcap_{i \leq n} \ker \varphi_i\right] \leq \prod_{i \leq n} [G : \ker \varphi_i] < \infty$. Lastly, $\{e\} = \bigcap_{n=1}^{\infty} \ker(\varphi_n)$, since an element $a_n \neq e$ always has $a_n \notin \ker(\varphi_n)$.

($\Leftarrow$) Suppose $G \geq G_1 \geq G_2 \geq \cdots$ are of finite index with $\bigcap_{n \geq 1} G_n = \{e\}$. Let $s \in G \setminus \{e\}$. Since $\bigcap_{n \geq 1} G_n = \{e\}$, there exists $n \geq 1$ such that $s \notin G_n$. By the previous corollary, there exists $\varphi \colon G \twoheadrightarrow G/K$ with $\varphi(s) \neq e$. □

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2.8 Alternative Proof that F_2 is Residually Finite

Consider the map $S \to \mathrm{GL}_2(\mathbb{Z})$ that extends

$$a \longmapsto \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad b \longmapsto \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

to a unique homomorphism $\varphi \colon F_2 \to \mathrm{GL}_2(\mathbb{Z})$. In fact, φ is injective by the Ping-Pong lemma (to come). Then apply the following:

Theorem (Malcev, 1965)

Every finitely generated subgroup of $\mathrm{GL}_n(\mathbb{F})$ is residually finite. □

Even this theorem is overkill. Let $w \in F_2$, $\varphi(w) \in \mathrm{GL}_2(\mathbb{Z})$. Pick $m \in \mathbb{Z}$ such that some entries in $\varphi(w)$ are not those of I modulo m. Then the entrywise modulo operation is a group homomorphism

$$\varphi \colon \mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z}), \qquad (a_{ij}) \longmapsto (a_{ij} \bmod m),$$

sending $\varphi(w)$ to some element other than the identity. Residual finiteness follows. □

Theorem

Every finitely generated residually finite group is Hopfian.

Proof. Let $f \colon G \twoheadrightarrow G$ be a surjective homomorphism. Let $s \in G$, $s \neq e$. Since G is residually finite, there exists $\pi \colon G \to H$ with $|H| < \infty$ and

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$\pi(s) \neq e$. Since G is finitely generated, $|\mathrm{Hom}(G, H)| < \infty$. We can construct homomorphisms $\pi \circ f^m \colon G \to H$ for all $m \in \mathbb{N}$. By pigeonhole, there exist $n < m$ such that $\pi \circ f^m = \pi \circ f^n$. Since f is surjective, we can cancel f^n: for $t = f^n(s)$,

$$\implies \pi(t) = \pi \circ f^n(s) = \pi \circ f^m(s) = \pi \circ f^{m-n}(t).$$

Thus $\pi \circ \underbrace{f \circ \cdots \circ f}_{m-n}(s) = \pi(s) \neq e$ by assumption. □

Corollary

F_n is Hopfian, since it is finitely generated and residually finite; i.e. there is no proper quotient of F_n isomorphic to it. □

Remark. For F_n and $\mathbb{Z}^n$, an explicit sequence exhibiting residual finiteness (p. 21) is $G \geq G^2 \geq G^4 \geq G^8 \geq \cdots$, where $G^n := \langle s^n \mid s \in G \rangle$. □

Proposition (Baumslag–Solitar group)

$\mathrm{BS}(2,3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$ is finitely generated but not Hopfian, hence not residually finite. [Proof deferred to later.] □

2.9 F_2, the Free Group on $\{a, b\}$

Question: Do $\{a, b^2\}$ or $\{ab, ba\}$ generate F_2?

Exercise

$\{a, b^2\}$ does not generate F_2.

Solution 1. $b \notin \langle a, b^2 \rangle$, by looking at word length

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(argue by induction). □

Solution 2. Consider the unique (by the universal property) homomorphism $f\colon F_2 \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$, $a \mapsto 0$, $b \mapsto 1$. Then $f(a) = f(b^2) = 0$, so $\langle a, b^2 \rangle \subseteq \ker f \neq F_2$. □

Remark. With this, we can even see that the normal subgroup generated by a, b^2 (considering all possible conjugates of $\langle a, b^2 \rangle$) is not the whole F_2. □

Exercise (Algebra 3)

$\{m_1, m_2, \ldots, m_n\}$ spans $\mathbb{Z}^n \iff$ the matrix given by $(m_1, m_2, \ldots, m_n)$ as column vectors is invertible in $M_n(\mathbb{Z}) \iff \det(m_1, m_2, \ldots, m_n) = \pm 1$. [Hint: matrices $\longleftrightarrow$ homomorphisms.]

Solution 3. We can also consider $f\colon F_2 \twoheadrightarrow \mathbb{Z}^2$, $a \mapsto \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$, $b \mapsto \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$, the abelianization map of F_2. Then f takes $S = \{a, b^2\}$ into the matrix $\left(\begin{smallmatrix}1&0\\0&2\end{smallmatrix}\right)$. This matrix is not invertible, hence $f(a), f(b^2)$ do not even generate $\mathbb{Z}^2$. □

Exercise

$\langle ab, ba \rangle \neq F_2$.

Solution. Again with the abelianization $F_2 \twoheadrightarrow \mathbb{Z}^2$: f brings $\langle ab, ba \rangle$ to the matrix $\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, which is again

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non-invertible, hence $f(ab), f(ba)$ do not even generate the image $\mathbb{Z}^2$. □

Exercise

$S = \{a,\ baba^{-1}b^{-1}\}$ does not generate F_2.

Solution. Any map $f\colon \langle S\rangle \twoheadrightarrow$ an abelian group will not justify this, since $aba^{-1}b^{-1}$ becomes trivial, leaving the images as $f(a), f(b)$. Consider instead

$$F_2 \twoheadrightarrow S_3, \qquad a \mapsto (1\,2) = \sigma, \quad b \mapsto (2\,3) = \tau.$$

Then $(\tau\sigma)^3 = (\sigma\tau)^3 = e \implies f(baba^{-1}b^{-1}) = f(a) = \sigma$ by computation, hence the images do not generate S_3 $\implies$ S does not generate F_2. □

Theorem (Nielsen)

Any finite set S in a free group can be transformed in finitely many steps into a finite set T that forms a basis of $\langle S\rangle$. In particular, $\langle S\rangle$ is a free group, for any $|S| < \infty$. In other words, every subgroup of a free group is free. □

3 Group Presentations — Definitions and Consequences

Definition

Let G be a group and $S \subseteq G$. The normal closure / normal subgroup generated by S is

$$\langle\langle S\rangle\rangle_G := \bigcap \{N \trianglelefteq G : S \subseteq N\}.$$

Concretely,

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$$\langle\langle S\rangle\rangle_G = \left\{g_1 s_1^{e_1} g_1^{-1} \cdots g_k s_k^{e_k} g_k^{-1} : k \geq 0,\ g_i \in G,\ s_i \in S,\ e_i = \pm 1\right\},$$

the set of all finite products of conjugates of elements of $S^{\pm 1}$. □

Remark. $\langle S\rangle_G \leq \langle\langle S\rangle\rangle_G$, with equality $\iff$ $\langle S\rangle_G \trianglelefteq G$. □

Definition (Presentations)

Let X be a set and $R \subseteq F_X$ a set of reduced words, called the relators. The group presented by $\langle X \mid R\rangle$ is

$$\langle X \mid R\rangle := F_X / \langle\langle R\rangle\rangle_{F_X}.$$

A relator r is sometimes written as "$r = e$"; a relation "$u = v$" means the relator uv^{-1}. A presentation of a group G is an isomorphism $G \cong \langle X \mid R\rangle$. G is finitely presented if it has a presentation with $|X|, |R| < \infty$. □

Example

$$\begin{aligned} F_n &= \langle x_1, \ldots, x_n \mid \varnothing\rangle, \\ \mathbb{Z}/n\mathbb{Z} &= \langle a \mid a^n\rangle = \langle a \mid a^n = e\rangle, \\ \mathbb{Z}^2 &= \langle a, b \mid aba^{-1}b^{-1}\rangle = \langle a, b \mid aba^{-1}b^{-1} = e\rangle, \\ \mathrm{BS}(p,q) &= \langle a, t \mid ta^p t^{-1} = a^q\rangle = \langle a, t \mid ta^p t^{-1} a^{-q}\rangle. \end{aligned}$$

□

Proposition

$w, w' \in F_X$ represent the same element of $\langle X \mid R\rangle \iff w'$ can be obtained from w by finitely many insertions or deletions of subwords of the form ss^{-1}, $s^{-1}s$, or $r^{\pm 1}$, where $s \in X$, $r \in R$. □

Proposition (Universal property of presentations)

Let $G = \langle X \mid R\rangle$, H a group, and $f_0 \colon X \to H$ a map whose extension $\tilde{f} \colon F_X \to H$ satisfies $\tilde{f}(r) = e$ for all $r \in R$. Then there exists a unique homomorphism $f \colon G \to H$ with $f(x) = f_0(x)$ for all $x \in X$. (Proof on p. 38.) □

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Example

$S = \{ab^{-1},\ a^{-1}b^2\}$ gives a group

$$F_2/\langle\langle S\rangle\rangle = \langle a, b \mid ab^{-1} = e,\ a^{-1}b^2 = e\rangle.$$

□

In fact, S generates a strict subgroup isomorphic to F_2. Yet no abelianization can show $\langle S\rangle \neq F_2$: the normal subgroup generated by S is $\langle\langle S\rangle\rangle = F_2$.

Solution. To prove this, Nielsen's idea is to look at products: let $u = ab^{-1}$, $v = a^{-1}b^2$. Then

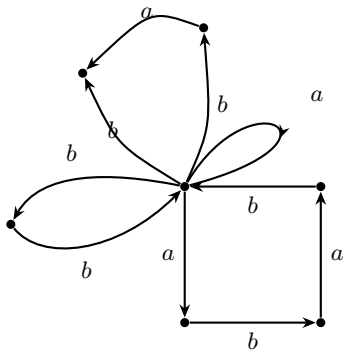
$$\left.\begin{aligned} u^2 &= ab^{-1}ab^{-1} \\ uv &= ab^{-1}a^{-1}b^2 \\ vu &= \cdots \\ &\vdots \end{aligned}\right\} \quad \text{length} \geq 2k.$$

Remark: the formalization of this is an induction on word length.
Hence $a, b \notin \langle ab^{-1}, a^{-1}b^2\rangle$. □

Exercise (Nielsen's algorithm / Stallings' folding argument)

Find a basis of $\langle a,\ bab^{-1},\ b^2,\ abab\rangle$.

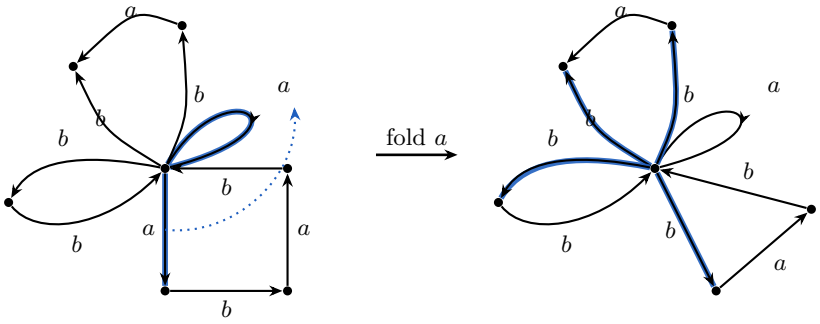
Step 1. Write the words as loops.



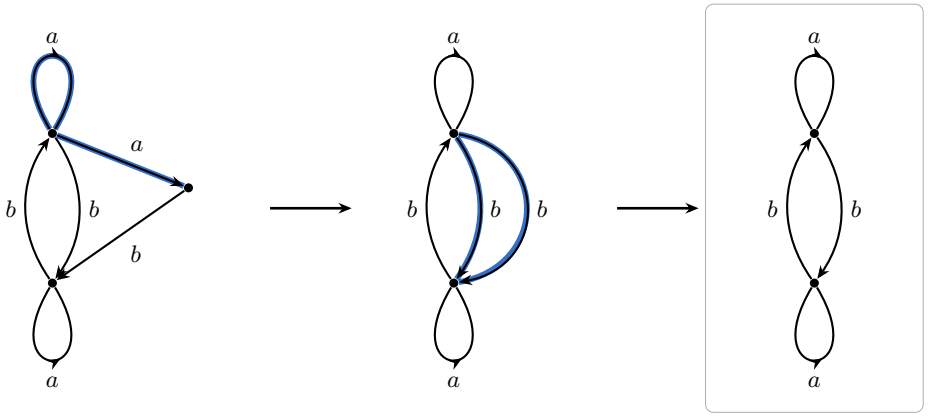
Step 2. Fold the same edges, so that each

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node has at most one out-edge / in-edge for each s_i. First fold the two out-a-edges at the base vertex:



Now fold b: the four (blue) out-b-edges at the base vertex are identified, then the two out-a-edges at the new vertex, and finally the two remaining out-b-edges at the base:



Step 3. Identify the fundamental cycles.

Definition (Fundamental cycles)

Let $G = (V, E)$ be a connected graph. Fix a base vertex $v_0 \in V$ and a spanning tree (ST) T rooted at v_0. A tree on V vertices has $V - 1$ edges, so there are $E - (V - 1) = E - V + 1$ edges not in T. For every such edge e there is a closed walk

$$v_0 \xrightarrow{\text{in } T} \text{start of the edge } e \longrightarrow \text{end of the edge } e \xrightarrow{\text{in } T} \text{back to } v_0,$$

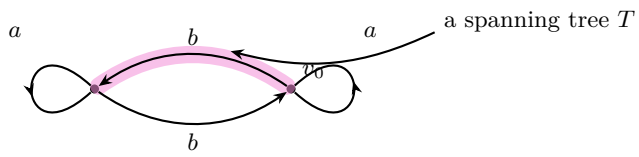
where $\xrightarrow{\text{in } T}$ means "inside the spanning tree". □

Then, counting the fundamental cycles, we get a basis of $\langle S \rangle$:

$$B_1 = \{a,\ b^2,\ b^{-1}ab\} \qquad \text{or} \qquad B_2 = \{a,\ b^2,\ bab^{-1}\},$$

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depending on the choice of spanning tree T:



Indeed, $\pi_1(G, v_0)$ = closed walks at v_0 modulo backtracking, which is free on the fundamental cycles (see page 31 below); meanwhile, all the generators can be recovered from the folded graph. □

Remark. $\langle S\rangle = \langle T\rangle$ is the kernel of

$$F_2 \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad a \mapsto 0, \quad b \mapsto 1.$$

$\langle S\rangle$ is an index-2 subgroup of rank 3 inside F_2. In fact, the folded graph of $\langle S\rangle$ is exactly $K\backslash \mathrm{Cay}(G, S)$, taking $G = F_S/K = \mathbb{Z}/2\mathbb{Z}$ (see p. 33 below). □

Remark. In general, $\langle S\rangle$ has finite index $\iff$ the folded graph of S is complete: for every letter a, every node has an in-a-edge and an out-a-edge. If so, the index of $\langle S\rangle$ = the number of vertices in the folded graph. □

Remark. A free group of rank 2 (F_2) contains a group of rank 3: $\langle S\rangle$ with $S = \{a, b^2, b^{-1}ab\}$. □

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3.1 Presentations of Free Groups

Theorem (Nielsen–Schreier)

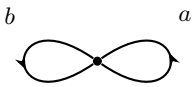
Any subgroup of a free group is free.

Theorem (Nielsen–Schreier index)

If $G \subseteq F_n$ is a finite index subgroup of index N, then

$$\mathrm{rk}(G) - 1 = N\big(\mathrm{rk}(F_n) - 1\big).$$

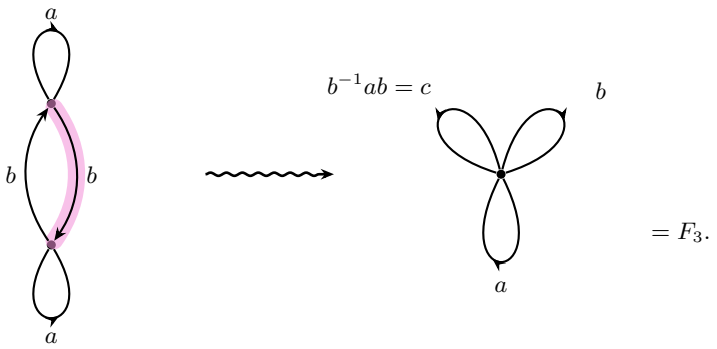
Sketch of proof. Let $X = \mathrm{Cay}(F_2, S)$. $B_2 = F_2\backslash X$ is the quotient graph:



Let $G \leq F_2$. Consider $G\backslash X =: X'$. Then $G = \pi_1(X')$ (covering space theorem / see below, p. 31).

Claim: The fundamental group of a graph is a free group.

The fundamental group = loops on the graph, classified up to continuous deformations of the loop. If X' is a graph, then X' contains a maximal subtree. Since π_1 is invariant under deformation, contract the tree to a point; the graph ends up a bouquet of cycles. E.g.:

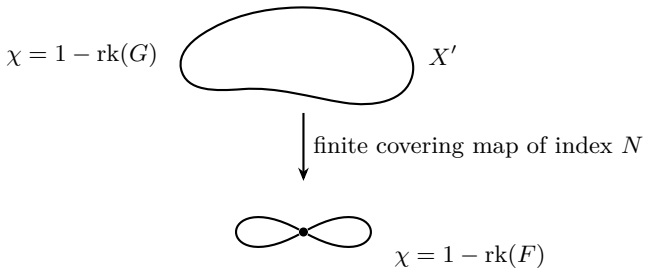


□

Sketch of proof (Nielsen–Schreier index). Source manuscript page 31

Consider $X' \to B_2$ in the previous proof. Let χ = the Euler characteristic $= V - E + \text{Faces} = V - E$.

(1) χ is a homotopy-equivalence (i.e. continuous deformation) invariant.
(2) χ is multiplicative under finite covers.



By (2), $1 - \text{rk}(G) = N\big(1 - \text{rk}(F)\big)$.

Remark. We can restate the entire proof with no algebraic topology at all, but only graph and group theory: $X' = G\backslash X$ has $V =$ the cosets Gx, hence $V = [F_2 : G] = N$. Every vertex has exactly one out-edge and one in-edge per generator (given finite index), thus $E = nN$.

Theorem (covering spaces). $G = \pi_1(X')$: a walk in X' starting at $v_0 := Ge$ ends at Gw after reading w. $Gw = Ge \iff w \in G$. Hence

$$\pi_1(X') = \text{loops modulo continuous deformations} = \text{loops modulo backtracking} = G.$$

Lastly, by counting fundamental cycles (pp. 28–29), there are exactly $E - V + 1 = 1 - \chi$ of them. Thus $\text{rk}(G) = \#$ fundamental cycles in (V, E), which generate G, $= nN - N + 1$, i.e.

$$\text{rk}(G) - 1 = N\big(\text{rk}(F_n) - 1\big).$$

□

✠ Note. These results together give a deterministic way to

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test if a set generates a free group: run Stallings' folding on it. If the folded graph is incomplete, then "no" (remark on p. 29). If complete, then S is a generating set $\iff \text{rk}(\langle S \rangle) = \text{rk}(F_n)$ (by the Nielsen–Schreier index), where $\text{rk}(\langle S \rangle)$ can be read off from the folded graph. □

3.2 Nielsen's Elementary Transformations

Nielsen's elementary transformations on a set $S = \{s_1, \dots, s_n\}$ spanning F_n:

$$\left.\begin{array}{l}(1)\text{ permute } s_i \text{ and } s_j;\\ (2)\text{ change } s_i \text{ to } s_i^{-1};\\ (3)\text{ change } s_i \text{ to } s_i s_j,\ j \neq i.\end{array}\right\} \in \mathrm{Aut}(F_n)$$

Theorem (Nielsen)

These three transformations generate $\mathrm{Aut}(F_n)$. □

Remark. This shows that for another free generating set $T \subseteq F_n$, we can get $S \longleftrightarrow T$ in finitely many Nielsen elementary transformations. □

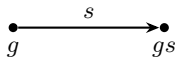
Theorem

For a normal subgroup $G \trianglelefteq F_S$,

$$G\backslash \mathrm{Cay}(F_S, S) \;\cong\; \mathrm{Cay}(F_S/G, S).$$

In particular, if G is of finite index, then these are furthermore graph-isomorphic to the folded graph of G from Stallings' folding argument.

"Proof". The vertex set of the quotient graph is exactly F_S/G by definition. Both edge sets are (vertex, letter) pairs (g, s):



(identified).

Every edge with such a label

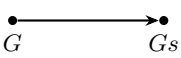
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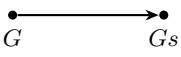
ends at the same place:

$$\begin{aligned}&\text{in } \mathrm{Cay}(F_S/G, S): \quad (Gw)s := (Gw)(Gs) = Gws;\\ &\text{in } G\backslash \mathrm{Cay}(F_S, S): \quad (Gw)s = Gws.\end{aligned}$$

[Really, the S in $\mathrm{Cay}(F_S/G, S)$ represents $\{Gs : s \in S\}$.]
Hence they have identical $V, E \implies$ they are the same.

If G has finite index, then its folded graph has exactly one edge out and one edge in for every generator. The folded graph has each generator as a loop at e. Hence every element in G becomes trivial on the graph, as it will always walk to e. This is the same as F_S/G. Furthermore, the edge labelled s does:

in $\mathrm{Cay}(F_S/G, S)$: 

in the folded graph: 

($e \mapsto s$ on the graph, but everything mod G).

Thus the folded graph of G is $\mathrm{Cay}(F_S/G, S)$. □

3.3 Recipe to Build Any Group

(1) Start with F_S.
(2) Choose $R =$ a set of words (relators).
(3) $G = F_S/\langle\langle R\rangle\rangle$, where $\langle\langle R\rangle\rangle$ is the normal closure of R.

Fact

Let G be a group and $G = \langle S\rangle$ with S a generating set. Then

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there exists a surjective homomorphism $F_S \twoheadrightarrow G$. Choose a generating set R of the kernel as a normal closure (i.e. $\langle\langle R\rangle\rangle = \ker$); then (S, R) is called a presentation of G, denoted by $G = \langle S \mid R\rangle$. □

Example

See p. 26. □

Example (Klein's four group K_4)

$$K_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \langle a, b \mid a^2 = b^2 = e,\ ab = ba\rangle = F_2/\langle\langle a^2,\ b^2,\ aba^{-1}b^{-1}\rangle\rangle,$$

where the normal subgroup is the kernel of the homomorphism

$$\varphi\colon F_2 \xrightarrow[\text{abelianization}]{} \mathbb{Z}^2 \xrightarrow[\text{mod } 2]{} (\mathbb{Z}/2\mathbb{Z})^2,$$

and $K_4 = (\mathbb{Z}/2\mathbb{Z})^2 = H_1(F_2, \mathbb{Z}/2\mathbb{Z})$, the homology of F_2 mod 2. Indeed, φ kills every relator, hence $\ker\varphi \supseteq \langle\langle R\rangle\rangle$. Then there is a well-defined surjective map

$$\tilde{\varphi}\colon F_2/\langle\langle R\rangle\rangle \twoheadrightarrow F_2/\ker\varphi \xrightarrow{\sim} (\mathbb{Z}/2\mathbb{Z})^2.$$

Hence $|F_2/\langle\langle R\rangle\rangle| \geq |(\mathbb{Z}/2\mathbb{Z})^2| = 4$. On the other hand, notice that $F_2/\langle\langle R\rangle\rangle$ has size 4: every element can be reduced to one of $\{e, a, b, ab\}$. Therefore it must be that $\langle\langle R\rangle\rangle = \ker\varphi$. Moreover, by the Nielsen–Schreier theorem and the Nielsen–Schreier index, $\langle\langle R\rangle\rangle$ is free and it has rank

$$\operatorname{rk}(\langle\langle R\rangle\rangle) = N\big(\operatorname{rk}(G) - 1\big) + 1 = 4 \cdot (2 - 1) + 1 = 5.$$

□

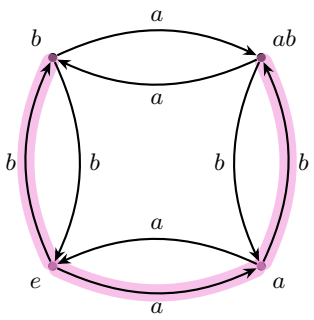
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Example

Compute a basis for $\langle\langle R\rangle\rangle$ in the K_4 example. □

Solution. We don't know what the generators of $\langle\langle R\rangle\rangle$ are: R only generates it as a normal closure. Yet $\langle\langle R\rangle\rangle$ is of finite index in F_2, hence, by the previous theorem, its folded graph is

$$\operatorname{Cay}\big(F_2/\langle\langle R\rangle\rangle, \{a, b\}\big) = \operatorname{Cay}\big((\mathbb{Z}/2\mathbb{Z})^2, \{a, b\}\big) =$$



Pick the base point e; a spanning tree is . Then for each edge not in the spanning tree, we have a closed loop from e touching it, i.e. the fundamental cycles:

$$b^2, \quad a^2, \quad ab^2a^{-1}, \quad abab^{-1}, \quad bab^{-1}a^{-1}.$$

This is a basis of $\langle\langle R \rangle\rangle$. □

Example

Removing the relator $ab = ba$ from the Klein four group, we get the infinite dihedral group

$$D_\infty := \langle a, b \mid a^2,\ b^2 \rangle.$$

We can view a, b as reflections on $\mathbb{Z}$:

$$a \longmapsto (x \mapsto -x), \qquad b \longmapsto (x \mapsto 1 - x).$$

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ab has infinite order in D_∞, and the words $a, ab, aba, abab, \dots$ are all distinct. We have $D_\infty \twoheadrightarrow (\mathbb{Z}/2\mathbb{Z})^2$ by quotienting by $\langle aba^{-1}b^{-1} \rangle$, to get back K_4. □

Exercise

Prove that the group $\langle a, b \mid bab^{-1} = a^2,\ aba^{-1} = b^2 \rangle$ is trivial. [Hint: compute a few elements and see.] This group is H_2 ("Higman").

Proof. Note that

$$\begin{aligned} a^4 = (bab^{-1})^2 = ba^2b^{-1} = bbab^{-1}b^{-1} = b^2ab^{-2} = aba^{-1}\, a\, ab^{-1}a^{-1} \\ = abab^{-1}a^{-1} = aa^2a^{-1} = a^2. \end{aligned}$$

Thus $a^2 = e$. Moreover, $a^2 = e \implies bab^{-1} = e \implies ba = b \implies a = e$. Thus $b = e$. Thus the group is trivial. □

Exercise (Serre)

Prove that the group

$$\langle a, b, c \mid bab^{-1} = a^2,\ cbc^{-1} = b^2,\ aca^{-1} = c^2 \rangle = H_3$$

is trivial.

Proof. Idea: reduce c to a, b.

$$aca^{-1} = c^2 \implies ac^{-1}a^{-1} = c^{-2} \implies cac^{-1} = c^{-1}a.$$

Then, combined with $cbc^{-1} = b^2$, we get

$$b^2c^{-1}ab^{-2} = cbc^{-1}\,cac^{-1}\,cb^{-1}c^{-1} = cbab^{-1}c^{-1} = ca^2c^{-1} = cac^{-1}cac^{-1} = (c^{-1}a)^2.$$

Expanding, we get

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$$\begin{aligned} b^2c^{-1}ab^{-2} &= b^2c^{-1}b^{-2}\,b^2ab^{-2} = b^2c^{-1}b^{-2}\,ba^2b^{-1} \qquad \left[b^2cb^{-2} = cb^{-1} = b^{-2}c\right] \\ &= (b^2cb^{-2})^{-1}a^4 = (b^{-2}c)^{-1}a^4 = c^{-1}b^2a^4, \end{aligned}$$

while $(c^{-1}a)^2 = c^{-1}ac^{-1}a = c^{-1}c^{-2}aa = c^{-3}a^2$. Thus

$$c^{-3}a^2 = c^{-1}b^2a^4 \implies c^2 = a^{-2}b^{-2}.$$

On the other hand, $aca^{-1} = c^2 \implies aca^{-1} = a^{-2}b^{-2} \implies c = a^{-3}b^{-2}a$

$$\implies c^2 = a^{-3}b^{-2}a\,a^{-3}b^{-2}a = a^{-3}b^{-2}a^{-2}b^{-2}a.$$

$$\begin{aligned} &\implies a^{-2}b^{-2} = a^{-3}b^{-2}a^{-2}b^{-2}a \\ &\implies ab^{-2}a^{-1} = b^{-2}a^{-2}b^{-2} \qquad (\text{left} \times a^3,\ \text{right} \times a^{-1}) \\ &\implies ab^{-2}a^{-1} = (ab^{-1}a^{-1})^2 = (b^{-1}a)^2 = b^{-1}(ab^{-1})a = b^{-1}b^{-1}a^2a = b^{-2}a^3 \\ &\implies b^{-2}a^{-2}b^{-2} = b^{-2}a^3 \implies a^5 = b^{-2} \quad (\text{thus } a^{-5} = b^2). \end{aligned}$$

Then compute $a^{10} = (a^2)^5 = ba^5b^{-1} = bb^{-2}b^{-1} = b^{-2} = a^5$

$$\implies a^5 = e \implies b^2 = b^{-2} = e \implies a = b^2ab^{-2} = ba^2b^{-1} = a^4 \implies a^3 = e \implies a^{3+3-5} = a = e.$$

Then $b = e$, $c = e$ follow. □

3.4 Higman's Example

$$H_4 = \langle a, b, c, d \mid aba^{-1} = b^2,\ bcb^{-1} = c^2,\ cdc^{-1} = d^2,\ dad^{-1} = a^2 \rangle.$$

Theorem

H_4 is infinite with no nontrivial finite quotient, yet it admits a finite presentation. □

Proposition (Universal property of presentations)

Let $G = \langle X \mid R \rangle$, H a group, and $f_0 \colon X \to H$ a map whose extension $\tilde{f} \colon F_X \to H$ satisfies $\tilde{f}(r) = e$ for all $r \in R$. Then there exists a unique homomorphism $f \colon G \to H$ with $f(x) = f_0(x)$

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for all $x \in X$. We may denote $\tilde{f}(r)$ as $r\big(\tilde{f}(s_1), \ldots, \tilde{f}(s_n)\big)$, since the unique universal homomorphism replaces $s_i \mapsto f(s_i)$ for every word r in F_n, as ① it extends f and ② it preserves multiplication.

Proof.

$$G = F_X/\langle\langle R \rangle\rangle \xrightarrow[\text{natural projection}]{\pi} F_X/\ker(\tilde{f}) \xrightarrow[\text{first isomorphism theorem applied to } \tilde{f}]{\sim} H,$$

as $\langle\langle R \rangle\rangle \leq \ker(\tilde{f})$ from the assumption. Uniqueness is immediate, since two group homomorphisms agreeing on a generating set must agree with each other. □

Example

A homomorphism $\langle a, b \mid aba^{-1}b^{-1}\rangle \to H$ is the same thing as a commuting pair in H, i.e. there is a bijection

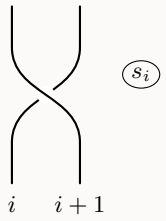
$$\{f\colon \langle a, b \mid aba^{-1}b^{-1}\rangle \to H\} \longleftrightarrow \{(x, y) \in H^2 \mid xy = yx\}.$$

Indeed, for every f, $(f(a), f(b))$ is a commuting pair $(x, y) \in H^2$. Conversely, given $x, y \in H$ with $xy = yx$, the universal property of presentations builds a unique homomorphism $f\colon \langle a, b \mid aba^{-1}b^{-1}\rangle \to H$. □

Example (braid groups)

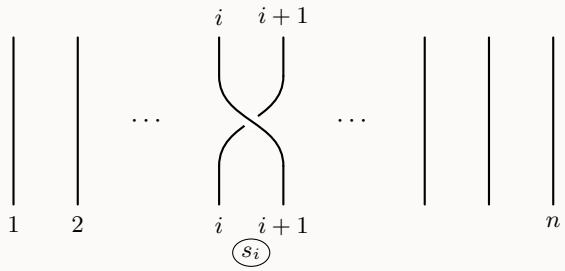
A classical relation: $aba = bab$, the braid relation. B_n = the braid group = braiding operations on n hanging strands. $s_i \in B_n$ is "strand i crosses in front of strand $i+1$":

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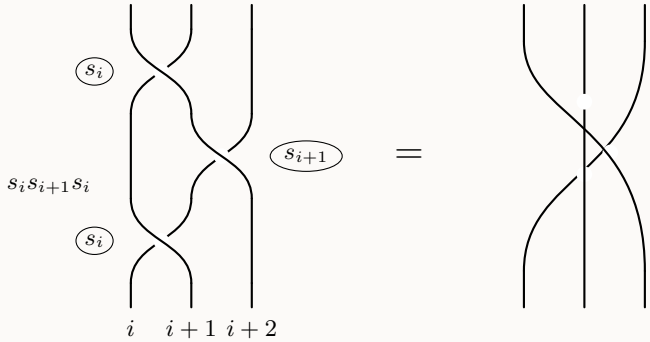


$$B_n = \left\langle s_1, \ldots, s_{n-1} \;\middle|\; s_i s_j = s_j s_i \ (|i-j| \geq 2),\quad s_{i+1}s_i s_{i+1} = s_i s_{i+1} s_i \right\rangle.$$

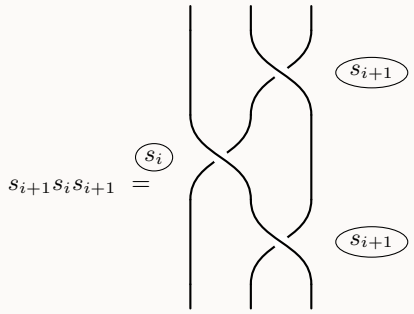
Each s_i is a braiding:



The braid relation $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$, read from top to bottom:



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□

Proposition

There exists a surjective homomorphism

$$f\colon B_n \twoheadrightarrow S_n, \qquad s_i \longmapsto (i,\, i+1).$$

Proof. Intuitively, we forget about which strand is in front of / under another strand, and only look at where each strand ends.

There is such a homomorphism: verify that the relators have image e,

$$f\left(s_is_{i+1}s_is_{i+1}^{-1}s_i^{-1}s_{i+1}^{-1}\right) = (i\ \ i{+}1)(i{+}1\ \ i{+}2)(i\ \ i{+}1)(i{+}1\ \ i{+}2)(i\ \ i{+}1)(i{+}1\ \ i{+}2) = e.$$

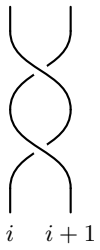
Hence there exists a unique homomorphism $B_n \to S_n$ by the universal property of group presentations.

Surjective: every permutation is a product of transpositions. Moreover, every transposition is a product of adjacent transpositions:

$$(i\ j) = (i\ \ i{+}1)(i{+}1\ \ i{+}2)\cdots(j{-}2\ \ j{-}1)(j{-}1\ \ j)(j{-}1\ \ j{-}2)(j{-}2\ \ j{-}3)\cdots(i\ \ i{+}1).$$

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It cannot be injective: $s_i^2 \mapsto e$, yet it is nontrivial:



□

Proposition

$$S_n = \left\langle t_1, \ldots, t_{n-1} \;\middle|\; t_i^2 = e,\ \ t_it_j = t_jt_i\ (|i-j| \geq 2),\ \ t_it_{i+1}t_i = t_{i+1}t_it_{i+1} \right\rangle,$$

i.e. the right-hand side is a presentation of S_n.

Proof. We have shown the existence of a surjective homomorphism. It is left to show $\#\langle\cdots\rangle = \#S_n \leq n!$. Let

$$T_n := \left\langle t_1, \ldots, t_{n-1} \;\middle|\; t_i^2,\ [t_i, t_j]\ (|i-j| \geq 2),\ t_i t_{i+1} t_i = t_{i+1} t_i t_{i+1} \right\rangle.$$

$\underline{n=2:}$ $T_2 = \langle t_1 \rangle = \{e, t_1\}$, $\#T_2 \leq 2$.

Let $n \geq 2$. Then let $T' \leq T_n$ be the subgroup generated by $t_1, \ldots, t_{n-2}$. Its generators satisfy the relators of T_{n-1} (and other relators), hence it is a quotient of T_{n-1}. Observe that right multiplication by t_j permutes the $\leq n$ cosets

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$$\{T',\ T't_{n-1},\ T't_{n-1}t_{n-2},\ \ldots,\ T't_{n-1}t_{n-2}\cdots t_1\}.$$

For instance, for all $\gamma \in T' \leq T_8$, case

1. $(\gamma t_7 t_6 t_5 t_4 t_3)t_1 = \gamma t_1 t_7 t_6 t_5 t_4 t_3 = \gamma' t_7 t_6 t_5 t_4 t_3 \in T' t_7 t_6 t_5 t_4 t_3$.
2. $(\gamma t_7 t_6 t_5 t_4 t_3)t_2 = \gamma t_7 t_6 t_5 t_4 t_3 t_2 \in T' t_7 \cdots t_2$.
3. $(\gamma t_7 t_6 t_5 t_4 t_3)t_3 = \gamma t_7 t_6 t_5 t_4 \in T' t_7 t_6 t_5 t_4$.
4. $(\gamma t_7 t_6 t_5 t_4 t_3)t_4 = \gamma t_7 t_6 t_5\, t_3 t_4 t_3 = \gamma t_3 t_7 t_6 t_5 t_4 t_3 \in T' t_7 t_6 t_5 t_4 t_3$.
5. $(\gamma t_7 t_6 t_5 t_4 t_3)t_5 = \gamma t_7 t_6 t_5 t_4 t_5 t_3 = \gamma t_7 t_6 t_4 t_5 t_4 t_3 = \gamma t_4 t_7 t_6 t_5 t_4 t_3 \in T' t_7 t_6 t_5 t_4 t_3$.

Note that we are not asserting there are n distinct cosets, but only $\leq n$ cosets. Since $e \in T'$, as each element is a product of $\{t_i\}^{\pm 1} = \{t_i\}$, the element itself is recorded at least once in the n cosets, since its action on e gives back itself. Thus

$$\#\left\langle t_1, \ldots, t_{n-1} \;\middle|\; t_i^2,\ [t_i, t_j]\ (|i-j| \geq 2),\ t_i t_{i+1} t_i = t_{i+1} t_i t_{i+1} \right\rangle \leq n!.$$

Thus the existence of a surjection implies that it has to be a bijection, i.e. an isomorphism. □

$\underline{\text{Remark.}}$ In general, a way to prove that a presentation gives a group is to have a surjection from the universal property, and give a counting bound on the presented

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group. □

3.5 Finite Generation vs. the Quotient's Cayley Graph

Proposition

$K = \langle\langle R \rangle\rangle$ is finitely generated $\iff$ $\mathrm{Cay}(G, S)$ is a tree *plus* finitely many edges, where $G = F_S/K$.

Proof. ($\Leftarrow$) Note that $\mathrm{Cay}(G, S)$ is always a connected graph, hence if $\underset{\text{tree}}{T} \cup \underset{\text{edges}}{E'} = \mathrm{Cay}(G, S)$, T contains all the nodes, hence is a spanning tree. Then since $K = \pi_1\big(\mathrm{Cay}(F_S/K, S)\big)$ (covering space theorem, p. 31), it follows that K is generated by the fundamental cycles in $\pi_1(\mathrm{Cay}(F_S/K, S))$, which are exactly the finitely many edges outside T in E'.

($\Rightarrow$)

$\underline{\text{Note.}}$ Both the Nielsen–Schreier index formula and the fact $\mathrm{Cay}(F_S/G, S) =$ the folded graph of G, for a finitely generated group G, only hold if G has finite index in F_S. Thus these two statements cannot be used here, since we only know that K is finitely generated. □

K is finitely generated $\implies$ there exists a free basis $\{g_1, \ldots, g_m\}$ of K by Nielsen's algorithm / Stallings' folding. Let T be a spanning tree of $\mathrm{Cay}(F_S/K, S)$, and

$$C_0 := \text{the non-tree (i.e. } \notin T\text{) edges appearing in } \{g_i\}_{i \in [m]},$$

$$C := \text{the non-tree edges } (= \text{chords}).$$

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Clearly $|C_0| < \infty$, since $\{g_i\}_{i=1}^m$ is finite and each g_i passes through finitely many edges.

Claim: $C \subseteq C_0$.

Suppose $g \in C$ but $g \notin C_0$. Then there is a cycle c at e through g. Since c is a cycle in $\mathrm{Cay}(F_S/K, S)$, $Kc = Ke$ in $F_S/K \implies c \in K$. Since $(g_i)_{i\in[m]}$ generates K, $c =$ a product of g_i's. Yet the reduction of a word does not create the new letter g in c. $\lightning$ Thus $C \subseteq C_0 \implies |C| < \infty \implies \mathrm{Cay}(F_S/K, S) =$ a spanning tree of it $\sqcup\, C\,(= C_0)$. □

Proposition

Let S be finite, $G = \langle S \mid R\rangle$, $K = \langle\langle R\rangle\rangle \leq F_S$. Then K is finitely generated as a group $\iff$ $K = \{e\}$ or G is finite, i.e. $|G| < \infty$.

Proof. ($\Leftarrow$) If $K = \{e\}$, it is clearly finitely generated. If G is finite, $[F_S : K] = |G| < \infty$, and by Nielsen–Schreier,

$$\mathrm{rk}(K) = 1 + N\big[\mathrm{rk}(F_S) - 1\big] = 1 + |G|\big(|S| - 1\big) < \infty.$$

A basis of size $\mathrm{rk}(K)$ finitely generates K.

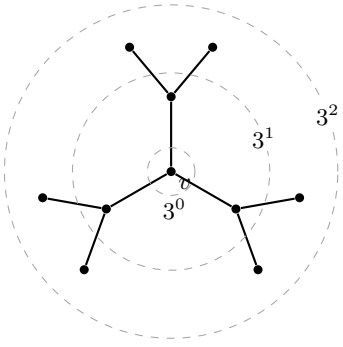
($\Rightarrow$) K finitely generated $\implies$ $\mathrm{Cay}(F_S/K, S)$ is a tree $\sqcup$ finitely many non-tree edges. Suppose, for a contradiction, that $K \neq \{e\}$ and $|G| = \infty$. Since $K \neq \{e\}$, there exists $r \in K \setminus \{e\}$, which is a cycle in $\mathrm{Cay}(F_S/K, S)$.

Claim: There exist $\{v_n\}_n$ with pairwise distance $> |r|$.

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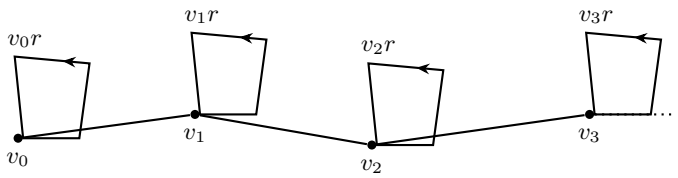
Induction on $n \in \mathbb{N}$. $n = 0$: pick a random vertex v_0. Suppose there exist $\{v_i\}_{i\in n}$ pairwise $|r|$-distant. Consider $B(v, k) := \{\text{vertices at distance} \leq k \text{ from } v\}$. Then, since at each vertex v there are at most $|S^{\pm 1}| = 2|S|$ vertices connected to it, for $i \in n$,

$$|B(v_i, |r|)| \leq \sum_{j=0}^{|r|} (2|S|)^j < \infty.$$



Thus $\left|\bigcup_{i=0}^{n-1} B(v_i, |r|)\right| < \infty$, i.e. it covers finitely many vertices of G. Since $|G| = \infty$, there exists $v_n \notin \bigcup_{i=0}^{n-1} B(v_i, |r|)$, i.e. v_n sits at distance $> |r|$ from $v_0, v_1, \ldots, v_{n-1}$.

By induction, we get an infinite sequence that is pairwise $|r|$-distant. Then notice that $\{v_n r\}_{n\in\mathbb{N}}$ is a sequence of disjoint cycles in $\mathrm{Cay}(F_S/K, S)$. Then this cannot be a tree $\sqcup$ finitely many chords, which only permits finitely many disjoint cycles:



$\lightning$ Thus $K = \{e\}$ or $|G| < \infty$.

Alternatively: The covering space theorem (p. 31) holds for any

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subgroup of a free group, i.e. $K = \pi_1\big(\mathrm{Cay}(F_S/K, S)\big)$. Suppose $|G| = \infty$, $K \neq \{e\}$. Then by the same disjoint cycles as above, there are infinitely many chords / non-tree edges $\implies$ an infinite basis consisting of the fundamental cycles through each chord.

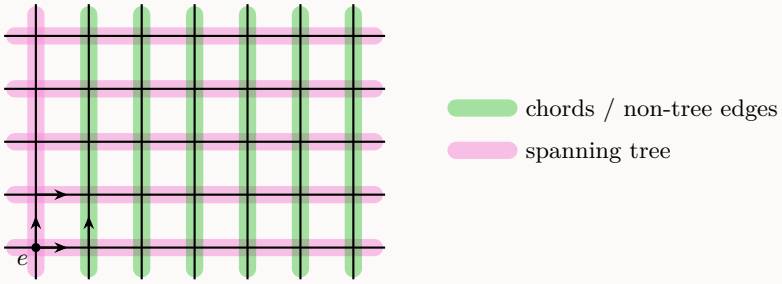
Claim: an infinite basis $\implies$ infinitely generated.

K has an infinite basis $\implies$ $K \twoheadrightarrow \bigoplus_{\text{chords}} \mathbb{Z}/2\mathbb{Z}$: as each chord is a fundamental cycle in K's basis, this map (pp. 15–16) sends $g \in K$ to $(a_c)_{c\in\text{Chords}}$, where a_c = the exponent sum of the "letter" / fundamental cycle c modulo 2 (this is the abelianization mod 2 map, i.e. $K \twoheadrightarrow K/K^2 \cong \bigoplus_{\text{Chords}}(\mathbb{Z}/2\mathbb{Z})$). As K is finitely generated, the generators pass through the map to generators of $\bigoplus_{\text{chords}}(\mathbb{Z}/2\mathbb{Z})$ $\implies$ $\bigoplus_{\text{chords}}(\mathbb{Z}/2\mathbb{Z})$ is a finite dimensional $\mathbb{Z}/2\mathbb{Z}$-vector space. But a finite set cannot generate the standard basis $(e_c)_{c\in\text{chords}}$, i.e. e_c has 0 on every component except on the component of c, where it has 1. $\lightning$ Thus either $|G| < \infty$ or $K = \{e\}$. $\square$

Example

$R = \{aba^{-1}b^{-1}\}$, $F_2/\langle\langle aba^{-1}b^{-1}\rangle\rangle = \mathbb{Z}^2$ (here $\langle\langle aba^{-1}b^{-1}\rangle\rangle = [F_2, F_2]$). $\mathrm{Cay}(\mathbb{Z}^2, \{a, b\})$:

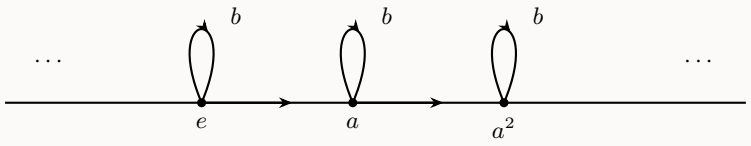
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$\boxed{R = aba^{-1}b^{-1}}$: infinitely (countably) many disjoint loops $\implies$ countably infinitely many fundamental cycles $\implies$ $\langle\langle R\rangle\rangle$ is free on countably infinitely many generators; we write $\langle\langle R\rangle\rangle \cong F_\infty$. $\square$

Example

$\mathbb{Z} = \langle a, b \mid b = e\rangle$, $K = \langle\langle b\rangle\rangle$.



$\mathrm{Cay}\big(F_2/K, \{a, b\}\big) = \mathrm{Cay}\big(\mathbb{Z}, \{a, b\}\big)$

A spanning tree is the axis, hence a collection of fundamental cycles $= \{a^n b a^{-n} : n \in \mathbb{Z}\}$. Thus K is free on $\{a^n b a^{-n} : n \in \mathbb{Z}\}$, $K \cong F_\infty$, and it is infinitely generated. □

Exercise

G is finitely generated and $G = \langle S \rangle$; then S contains a finite generating set.

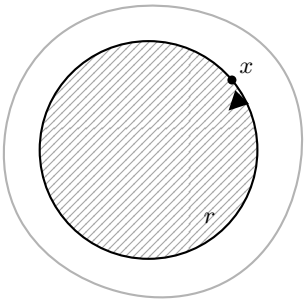
Solution. Let T be a finite generating set. For $t \in T$ there exists a word $w_t(s_1, \dots, s_n) = t$. Then $S_0 = \{\text{all } s_i \text{ for all } t \in T\}$, with $|S_0| < \infty$,

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generates G. □

3.6 Cayley 2-Complex — Adding Connectedness

Let $G = \langle S \mid R \rangle$, $X =$ the Cayley graph $= \mathrm{Cay}(G, S)$. $X' = \mathrm{Cay}(G, S, R)$, the Cayley 2-complex, is obtained by gluing in the relations (a 2-dimensional object). Every $r \in R$ gives a cycle. For every $x \in X$, every relation gives a cycle at x. Let $D(x, r)$ be the disk

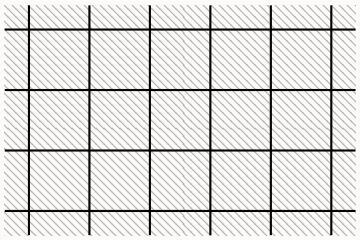


$= \text{an } n\text{-gon},$
$n = |r|$

$\mathrm{Cay}(G, S, R)$ is obtained from $\mathrm{Cay}(G, S)$ by gluing an n-gon $D(x, r)$ for all $x \in G$, $r \in R$.

Example

$\mathbb{R}^2 = \mathrm{Cay}\big(\mathbb{Z}^2, \{a, b\}, aba^{-1}b^{-1}\big)$:



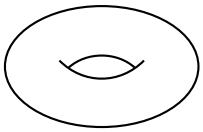
$= \mathbb{R}^2$

□

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$G \curvearrowright \mathrm{Cay}(G, S, R)$ (i.e. G acts on $\mathrm{Cay}(G, S, R)$): $h \in G$ sends g to hg, an edge ε to $h\varepsilon$, a disk $D(g, r)$ to $D(hg, r)$.

For $G = \mathbb{Z}^2$ and $\mathrm{Cay}(G, S, R) = \mathbb{R}^2$,

$$\underset{\mathbb{Z}^2}{G}\backslash\mathrm{Cay}(G, S, R) \cong$$  a torus.

Aside. Torus and quotient topology:

Definition

For a surjection $q\colon X \to Y$, we say $U \subseteq Y$ is open if $q^{-1}(U)$ is open in X. The topology of these open sets is called the quotient topology on Y. □

Define $x \sim y :\iff q(x) = q(y)$. On the other hand, if $\sim$ is a given equivalence relation, we define $q\colon X \to X/\sim$ via $q(x) = [x]$.

Lemma

A map out of a quotient $f\colon X/\sim \to Z$ is continuous $\iff$ $f \circ q$ is continuous. □

Lemma

A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. □

Proposition

Viewing $\mathbb{Z}^2$ as a normal subgroup of $\mathbb{R}^2$, the quotient $\mathbb{R}^2/\mathbb{Z}^2$ with the quotient topology is homeomorphic to $S^1 \times S^1$, where $S^1 = \{x + iy : x^2 + y^2 = 1\} \subseteq \mathbb{C}$.

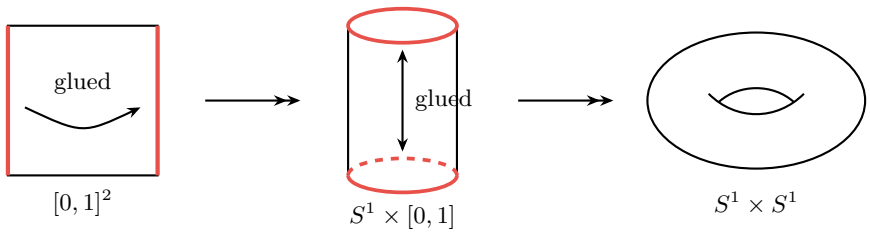
Proof. Define $F\colon \mathbb{R}^2 \to S^1 \times S^1$ via

$$(x, y) \longmapsto \left(e^{2\pi i x}, e^{2\pi i y}\right),$$

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this map is surjective and continuous. By Lemma 1, the induced $f\colon \mathbb{R}^2/\sim \hookrightarrow\!\!\!\!\rightarrow S^1 \times S^1$ is continuous. $S^1 \times S^1$ is a metric space, hence Hausdorff; notice that the quotient map $q\colon \mathbb{R}^2 \twoheadrightarrow \mathbb{R}^2/\mathbb{Z}^2$ is continuous by definition of the quotient topology, hence $\mathbb{R}^2/\mathbb{Z}^2 = q([0,1]^2)$, i.e. the continuous image of a compact set, hence compact. Hence by Lemma 2, f is a homeomorphism. Lastly, notice that $\mathbb{R}^2/\sim = \mathbb{R}^2/\mathbb{Z}^2$ as sets, hence they are equipped with the same quotient topology. Thus $\mathbb{R}^2/\mathbb{Z}^2 \cong S^1 \times S^1$. □

Remark. One way to realize this:



Indeed, one can show that $[0,1]^2/\sim$ is homeomorphic to $\mathbb{R}^2/\mathbb{Z}^2$. Applying the pair of lemmas twice, to the two surjections above, we have

$$\mathbb{R}^2/\mathbb{Z}^2 \;\cong\; [0,1]^2/\sim \;\cong\; S^1 \times S^1. \qquad \square$$

End of week 3.

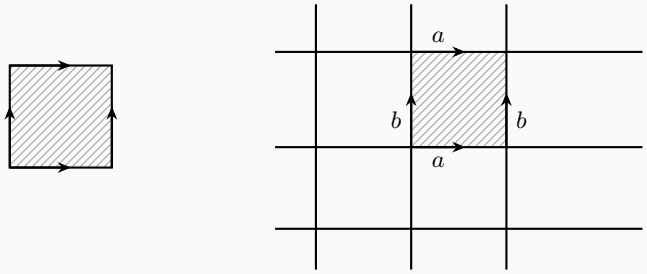
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4 Cayley 2-Complexes

Let $X = \mathrm{Cay}(G,S)$, G a group with $G = \langle S\rangle$. There exists the universal $f\colon F_S \twoheadrightarrow G$. Let R be a generating set of K as a normal subgroup, where $K =$ the relation kernel $= \ker f$.

Example (Cayley complex)

$r = aba^{-1}b^{-1}$, $\mathbb{Z}^2 = \langle a,b \mid aba^{-1}b^{-1}\rangle$. For $x \in \mathbb{Z}^2$, $D(x,r) =$ a disk at x on $r\colon x \mapsto x$, an n-gon with $n = |r|$:

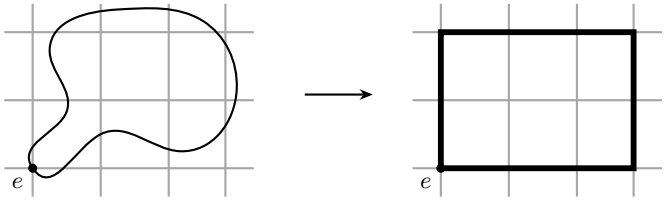


With $\mathrm{Cay}(\mathbb{Z}^2, \{a,b\})$, gluing all disks gives the Cayley 2-complex $= \mathbb{R}^2$. $\square$

Proposition

Let $G = \langle S \mid R\rangle$ be a presentation. Then the Cayley complex $X = \mathrm{Cay}(G,S,R)$ is simply connected, i.e. every loop can be contracted to a point, i.e. there are no holes.

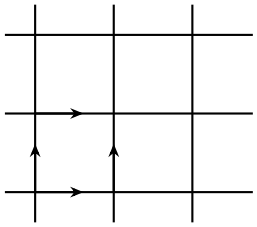
"Proof". A loop at e in X can be deformed to the 1-skeleton (i.e. $\mathrm{Cay}(G,S)$) into simplicial loops:



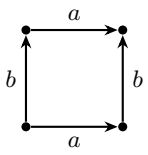
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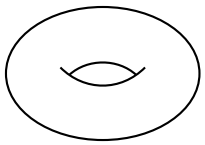
Then a loop is a word $w \in K$. Then, by definition of $K = \langle\langle R\rangle\rangle$, $w = \prod w_i^{-1} r_i w_i$ with $r_i \in R$. Each $w_i^{-1} r_i w_i$ is a contractible loop along the disk $D(w_i, r_i)$, hence their product is contractible. $\square$

Consider $V = \mathrm{Cay}(G,S,R)/G =:$ the presentation complex of G. The group acts on disks and vertices by translation.



$\mathrm{Cay}(G, S, R)$



$V =$  $= \mathbb{R}^2/\mathbb{Z}^2.$

Fact

$$G \xrightarrow{\text{drawing of}} V = \mathrm{Cay}(G, S, R)/G \xrightarrow{\pi_1} G.$$

The fundamental group recovers G. □

Example

$\mathrm{PSL}_2(\mathbb{Z}) = \langle s, t \mid s^2 = e,\ t^3 = e\rangle = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$, i.e. we identify A with $-A$. The presentation follows by sending

$$s \longmapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t \longmapsto \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}.$$

Indeed, $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)^{-1}\left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Then we can

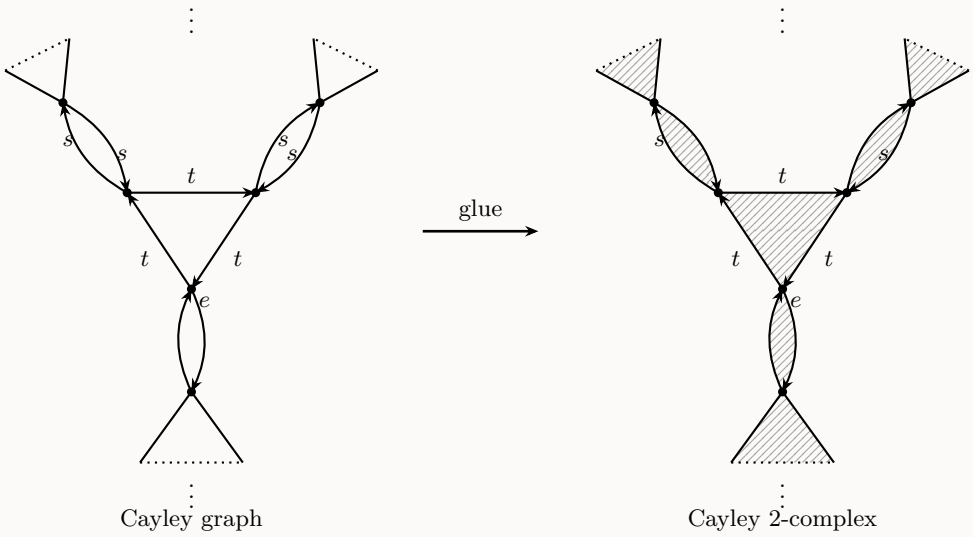
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always write an element of $\mathrm{PSL}_2(\mathbb{Z})$ as a product of $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = S$ and $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = T$. Observe $T^n = \left(\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right)$. Thus T^n adds n (R_2) to R_1. Take e.g.

$$\begin{aligned} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} &= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -3 & -2 \\ 2 & 1 \end{pmatrix} \qquad (\text{in } \mathrm{PSL}_2(\mathbb{Z})) \\ &= ST^{-2}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = ST^{-2}S\begin{pmatrix} -2 & -1 \\ 1 & 0 \end{pmatrix} \\ &= ST^{-2}ST^{-2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = ST^{-2}ST^{-2}S. \end{aligned}$$

Algorithm: keep doing Euclidean division on the first column entries by left-multiplying by S, T until one of the entries is 0. Hence $\mathrm{PSL}_2(\mathbb{Z}) = \langle S, T\rangle = \langle$images of s and $t\rangle$. Whether this map is injective, i.e. whether it has other nontrivial relators, requires the Ping-Pong lemma.

Now consider the Cayley graph of it, and glue in the disks to get the Cayley 2-complex:



□

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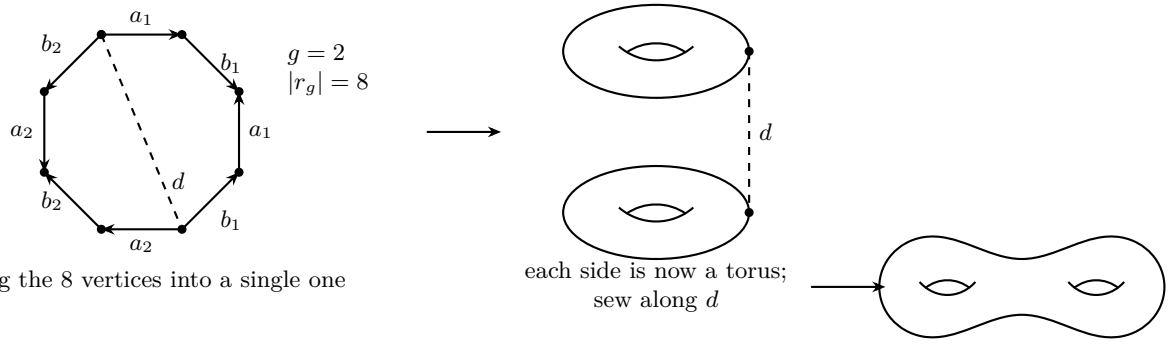
4.1 Presentation of the Surface Group ($g \geq 1$)

$$r_g = a_1 b_1 a_1^{-1} b_1^{-1}\, a_2 b_2 a_2^{-1} b_2^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

Recall, by definition of the presentation complex, that the presentation complex of the surface group

$$\Gamma_g := \langle a_1, b_1, \ldots, a_g, b_g \mid r_g \rangle$$

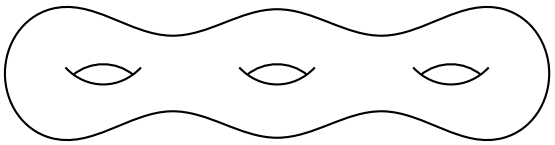
is defined by $\mathrm{Cay}\big(\Gamma_g, \{a_i, b_i\}, r_g\big)/\Gamma_g$, in which there is a single vertex, since Γ_g acts on the vertices Γ_g transitively; $2g$ edges, one for each generator; and one disk, for the single relator:



folding the 8 vertices into a single one

each side is now a torus; sew along d

In general, Γ_g's presentation complex is a genus-g surface, that is, g tori "sewn" together:



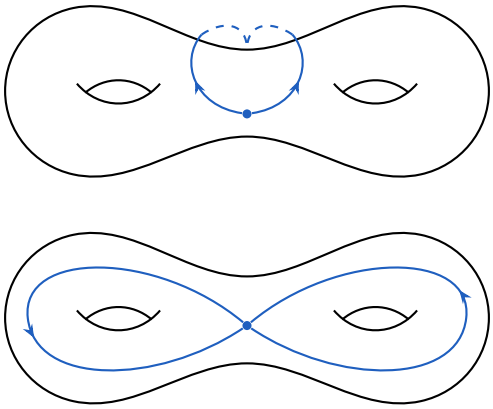
$g = 3$.

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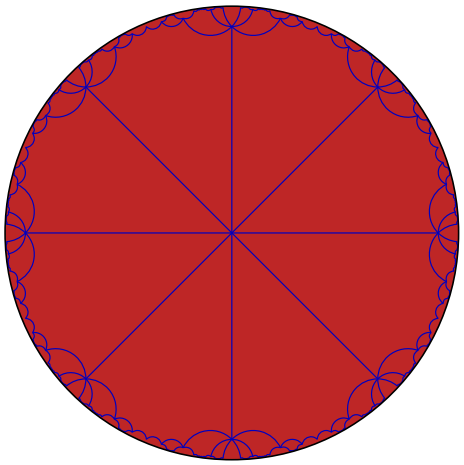
Thus, by the Fact on p. 52,

$$\boxed{\Gamma_g = \pi_1(\text{genus-}g \text{ surface})}$$

In particular, for $g = 2$, the $2g = 4$ edges can be identified on the surface as



Remark. The hyperbolic plane, tiled by octagons, is $\mathrm{Cay}(G, S, R)$ with $R = \langle\langle r_g \rangle\rangle$:



hyperbolic plane (tiled)
$= \mathrm{Cay}(G, S, R)$,
$R = \langle\langle r_g \rangle\rangle$

the $\{8, 8\}$ octagonal tiling

In fact, for $g \geq 2$, $\mathrm{Cay}\big(\Gamma_g, S, \langle\langle r_g \rangle\rangle\big)$ is

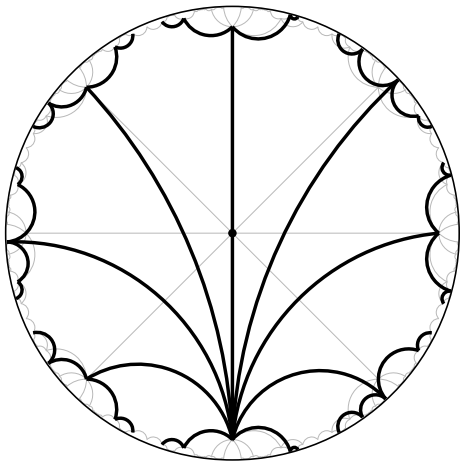
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realized as the hyperbolic plane tiled by regular $4g$-gons, which must have interior angles $\frac{2\pi}{4g}$, since each vertex has $4g$ edges connected to it. □

Proposition

For the group $\Gamma_g = \langle a_1, b_1, \ldots, a_g, b_g \mid r_g = e \rangle$, the subgroup of Γ_g generated by all but one generator is free.

"Proof" (idea). In the octagonally tiled hyperbolic plane, $\boxed{\text{find}}$ a tree on which the subgroup acts freely (i.e. $\mathrm{FP}_T(g) = 0$ for all $g \neq e$, where $|T^g| = \#$ fixed points of g in the tree T). Then apply the next proposition.



In the octagonally tiled hyperbolic plane, find a tree on which the subgroup acts freely (i.e. $\mathrm{FP}_T(g) = 0$, $\forall g \neq e$; $|T^g| = \#$ fixed points of g in the tree T).

□

Proposition

G is free $\iff$ G acts freely on a tree.

Proof. $G \curvearrowright X$ a tree $\implies$ $G \cong \pi_1(G\backslash X)$. $\pi_1(G\backslash X)$ is automatically free on its fundamental cycles. On the other hand, if G is free, then $G \curvearrowright \mathrm{Cay}(G, S)$ freely. □

Corollary (Nielsen–Schreier)

Subgroups of free groups are free.

Proof. A subgroup of a free group inherits the free group action, hence is free. □

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Example

Γ_g contains two free groups:

$$F_g^a = \langle a_1, \dots, a_g \rangle, \qquad F_g^b = \langle b_1, \dots, b_g \rangle.$$

They are free since they are subgroups of the free groups generated by $a_1, \dots, a_g, b_2, \dots, b_g$ (just removing one generator) and by $a_2, \dots, a_g, b_1, \dots, b_g$, respectively.

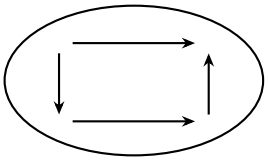
Claim: There exists a surjective homomorphism $\pi \colon \Gamma_g \twoheadrightarrow F_g^a \times F_g^b$.

Viewing it as a quotient group, we simply quotient by relations:

$$\begin{aligned} G &= \Gamma_g \big/ \langle\langle a_i b_j a_i^{-1} b_j^{-1} : i, j = 1, \dots, g \rangle\rangle \\ &= \langle a_1, b_1, \dots, a_g, b_g \mid \dots,\ a_i b_j = b_j a_i \ \forall i, j \rangle \\ &= \langle a_1, b_1, \dots, a_g, b_g \mid a_i b_j = b_j a_i \ \forall i, j \rangle \\ &\cong F_g^a \times F_g^b. \end{aligned}$$

□

Question: Are there other surjective homomorphisms $\Gamma_g \twoheadrightarrow F_g \times F_g$?
Answer: For all g, NO ($\iff$ the Poincaré conjecture is true). "Other" in the sense that it is not a conjugate homomorphism:



conjugate homomorphism

Exercise

Conjugacy classes in F_2.

Solution (sketch). Cyclically reduced words: conjugate $\iff$ rotation ($w_0 x \rightsquigarrow x w_0$).

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($\Leftarrow$) If so, $x\, w_0 x\, x^{-1} = x w_0$.

($\Rightarrow$) Conjugate: their axes both pass through e, and $uwu^{-1} = w_0 \implies \mathrm{Axis}(uwu^{-1}) = u\,\mathrm{Axis}(w)$. Note that

$$\mathrm{Axis}(w) = \bigsqcup_k [w^k, w^{k+1}] = \{w^k p : k \in \mathbb{Z},\ p \text{ a prefix string of } w\}.$$

Then $e \in u\,\mathrm{Axis}(w) \implies u^{-1} \in \mathrm{Axis}(w) \implies u^{-1} = w^k p$

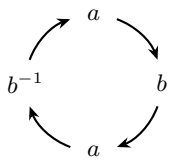
$$\implies uwu^{-1} = p^{-1} w^{-k} w w^k p = p^{-1} w p = p^{-1} p s\, p = sp,$$

where $w = ps$ with p a prefix and s the corresponding suffix string. Hence $uwu^{-1} = sp$ is a rotation of $w = ps$.

Every element is conjugate to a cyclically reduced word, and two cyclically reduced words are conjugate $\iff$ one is a rotation of the other. Thus

$$\{\text{conjugacy classes}\} \longleftrightarrow \{\text{necklaces of cyclically reduced representatives}\}.$$

For instance, the necklace of $abab^{-1}$ is



which is invariant under rotation of a word $ps \rightsquigarrow sp$, i.e. invariant under conjugacy of cyclically reduced words. Geometrically, conjugate $\implies$ same axis (as paths, not as $\{w^k\}$) or shifted axes; though this is not an identification, since a^2 and a have the same axis. □

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Exercise

$F_2 \twoheadrightarrow (\mathbb{Z}/2\mathbb{Z})^2$. Find a generating set of K.

Solution. See p. 35. □

Definition

For $m, n \in \mathbb{N}_{>0}$, we define the (free) Burnside group

$$B(m,n) := F_m / F_m^n.$$

□

Remark. It is not a free group. It is free in the sense that it is the largest group of exponent n (i.e. $g^n = e$ for all $g \in G$): by the universal property of presentations, every m-generated group of exponent dividing n is a quotient of $B(m,n)$. □

Exercise

Prove that the Burnside group with $n = 2$ is finite.

Solution. Recall we showed that for $n = 2$, $B(m,2) = F_m/F_m^2 \cong (\mathbb{Z}/2\mathbb{Z})^m$, hence finite. □

Exercise (harder)

Prove that $B(m,3)$ is finite.

Solution. Key: $uvu = v^{-1}u^{-1}v^{-1}$ [from $(uv)^3 = e$ in F_m/F_m^3]

$$\implies \boxed{xhx = h^{-1}x^{-1}h^{-1}, \qquad x^{-1}hx^{-1} = h^{-1}xh^{-1}, \qquad \text{and} \quad x^{-1}hx = (x^{-1}hx^{-1})x^2 = h^{-1}xh^{-1}x^{-1}.}$$

Induction on m: Base case: $B(1,3) = \mathbb{Z}/3\mathbb{Z}$ is finite.

Inductive step: let $B := B(m,3)$. Consider $H := \langle x_1, \dots, x_{m-1} \rangle \leq B$. By the universal property, since $H \leq B$ has exponent 3, it is a quotient of $B(m-1,3)$, hence by the inductive hypothesis

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$|H| \leq |B(m-1,3)| < \infty$. Now we bound what form an element of $B(m,3)$ can take.

Claim. Let $x := x_m$. Then

$$S = H \cup HxH \cup Hx^{-1}H \cup HxHx^{-1}H \cup Hx^{-1}HxH$$

is closed under right multiplication by every generator of $B = B(m,3)$.

Multiplication by an element of H is automatically closed in S. We show closure under multiplication by $x^{\pm 1}$:

- $Hx^{\pm 1} = Hx^{\pm 1}e \subseteq Hx^{\pm 1}H$;
- $Hx^{\pm 1}Hx^{\mp 1} = Hx^{\pm 1}Hx^{\mp 1}e \subseteq Hx^{\pm 1}Hx^{\mp 1}H$;
- $Hx^{\pm 1}Hx^{\pm 1} = HHx^{\mp 1}H = Hx^{\mp 1}H$;
- $HxHx^{-1}Hx = HxHHxHx^{-1} = HxHxHx^{-1} = HHx^{-1}HHx^{-1} = Hx^{-1}Hx^{-1} = HHxH = HxH$;
- $Hx^{-1}HxHx = Hx^{-1}HHx^{-1}H = Hx^{-1}Hx^{-1}H = HHxHH = HxH$.

Symmetrically, $HxHx^{-1}Hx^{-1} = Hx^{-1}H$ and $Hx^{-1}HxHx^{-1} = Hx^{-1}H$. Every element of $B(m,3)$ is a finite product of this form: $Hx^{\pm 1}Hx^{\pm 1}Hx^{\pm 1}\cdots x^{\pm 1}Hx^{\pm 1}H$. Thus $B(m,3) = S$

$$\implies |B(m,3)| \leq |H| + 2|H|^2 + 2|H|^3 \leq 5\,|B(m-1,3)|^3 < \infty.$$

□

End of Chapter 1.